

Supplementary Appendix

Do Firms Innovate When the Rivals They Watch Consolidate?

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1 The FactSet relationship data

The relationships used in this paper come from FactSet Revere Supply Chain Relationships, a data product that FactSet Research Systems distributes through its Standard DataFeed (FactSet Research Systems 2021*c*). The database is compiled by a team of analysts who read a listed company’s own public documents, including its annual reports and regulatory filings, investor presentations, press releases, and corporate website, and record the other companies that these documents identify as competitors, customers, suppliers, or strategic partners. The vendor describes the resulting data as a normalised classification of business relationships in four categories and presents them as a way of observing, around any listed company, the network of companies whose fortunes are tied to it. In FactSet’s vocabulary, each relationship is stored as a record containing a type, a *source* company (the company whose documents the analysts read), a *target* company (the company those documents identify), and two dates bounding the period over which the relationship was held to be current. A record without an end date is one that FactSet still regarded as current when the data were extracted (22% of all records are open). We treat such relationships as active through the extraction date and right-censor their duration at that date, rather than imputing persistence beyond the observed data window.

The data we hold run from 3 April 2003 to 23 November 2021. They contain 2,209,235 records covering 1,299,982 distinct directed pairs of companies and 16 relationship types, which Table 1 lists with their counts. Competitor records are the primary object of this paper; the remaining relationship types enter the analysis as channels through which a consolidation can reach a firm other than through its rivals.

*Wharton Research Data Services (WRDS) was used in preparing this Supplementary Appendix. This service and the data available thereon constitute valuable intellectual property and trade secrets of WRDS and/or its third-party suppliers.

Table 1: The relationship records by category and type, 2003–2021.

category	type	records	directed pairs	median length of a record (years)
Competitors	Competitor	414,858	305,850	1.6
Customers	Customer	879,797	541,696	1.0
Suppliers	Supplier	297,650	222,632	1.2
Strategic partners	Research collaboration	131,619	106,398	1.0
	Investor	128,928	101,406	1.2
	Equity investment	96,837	76,377	1.2
	Joint venture	69,329	45,416	1.3
	In-licensing	40,776	28,814	1.4
	Distribution	37,853	30,876	1.1
	Technology (legacy type)	31,511	26,251	1.2
	Out-licensing	25,089	18,537	1.3
	Marketing	23,505	18,725	1.1
	Manufacturing	11,859	9,404	1.4
	Licensing (legacy type)	7,796	7,201	1.1
	Integrated product offering	7,107	6,409	1.0
	Partner, type not identified	4,721	3,856	0.9
All		2,209,235	1,299,982	1.1

Figure 1 illustrates how the different types of records can gather around a single company. To construct the figure, we take a firm that is a source firm in 2015 and trace the relationships recorded around it in that year. The firm at the centre names eight competitors, which constitute its *designated rivals*; one additional firm names it as a competitor without being named in return. The second ring contains the suppliers, customers, and partners that its documents name. Because some of its designated rivals are themselves source firms in 2015, their own designated rivals are also observed and form the outer ring, the rivals of rivals. Every arrow points from the company whose documents name the relationship. The same structure, drawn around a consolidating firm rather than around the firm whose innovation we observe, provides the conceptual figure used in the main text.

The neighbourhood of one source firm in 2015: Vector Group Ltd. (US)

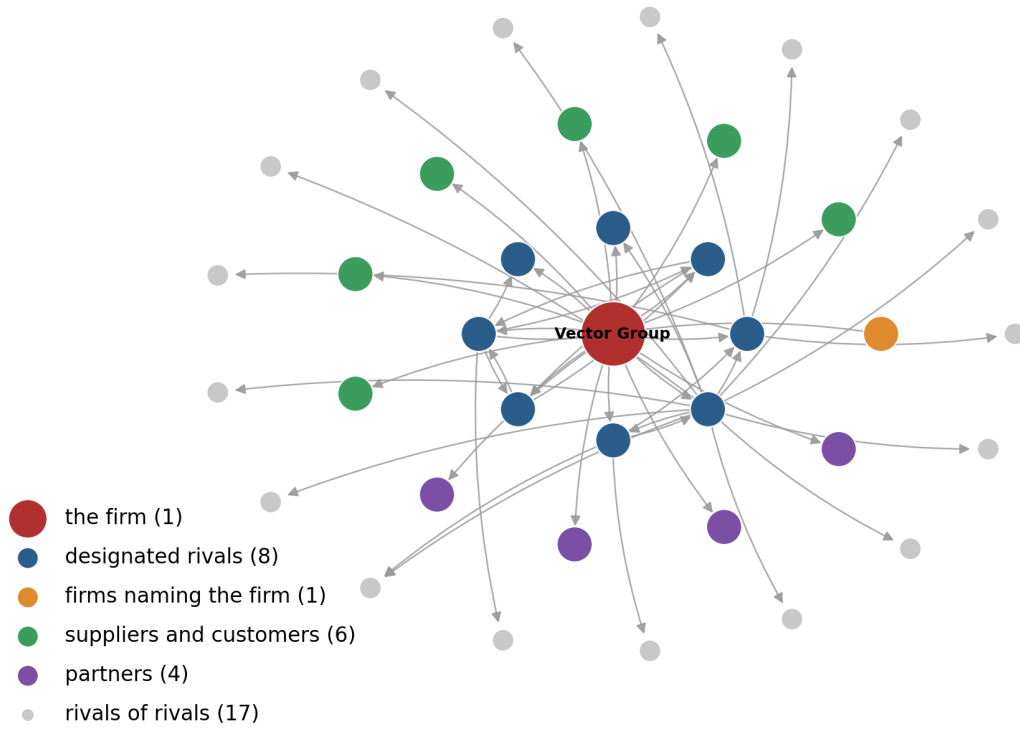


Figure 1: The 2015 neighbourhood of one source firm, drawn from the data: its designated rivals, the firm that designates it, its suppliers and customers, its partners, and the rivals of its rivals. Arrows point from the company whose documents identify the relationship.

The source firms

A company enters the data on the day the analysts first read its documents and publish a record from them, and from that day on it is a source firm. There are 41,490 source firms in 126 countries. Among them, 18,216 (44%) name at least one competitor, and these are the firms whose rival sets we observe and whose innovation the paper studies; the others report customers, suppliers or partners but no competitor, and they enter the analysis only through the relationships they report about the firms we study. Figure 2 counts the source firms with a live record in each calendar year, by the region of their country, and draws as a line the number of those with a live competitor record; Figure 3 maps the source firms by country, and Figure 4 dates their entry by the year of their first record. The eight largest countries are the United States (9,864), China (4,325), Japan (3,461), the United Kingdom (1,897), South Korea (1,544), Canada (1,405), Australia (1,129), Hong Kong (1,070).

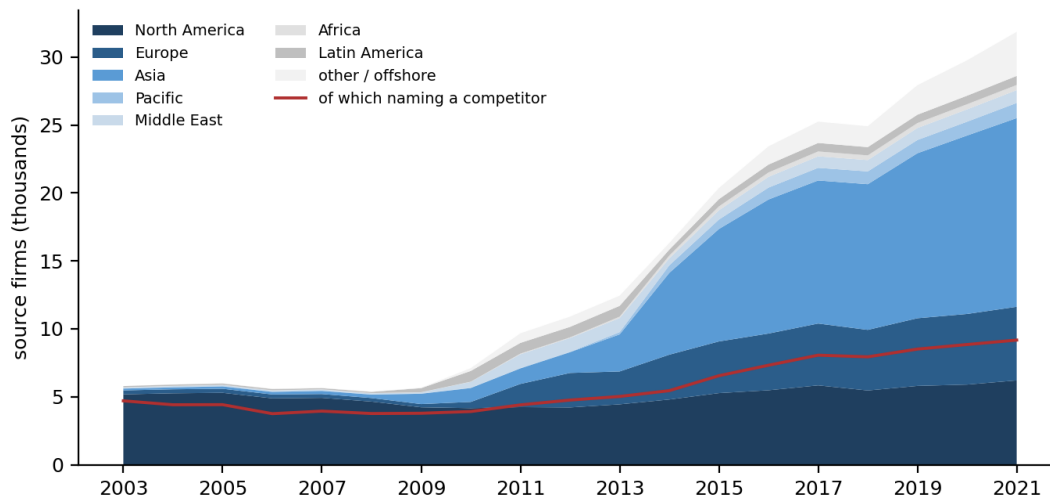


Figure 2: Source firms with a live record of any type in each calendar year, by the region of the firm's country, and among them the firms with a live competitor record (line).

Source firms (the source of at least one record), by country

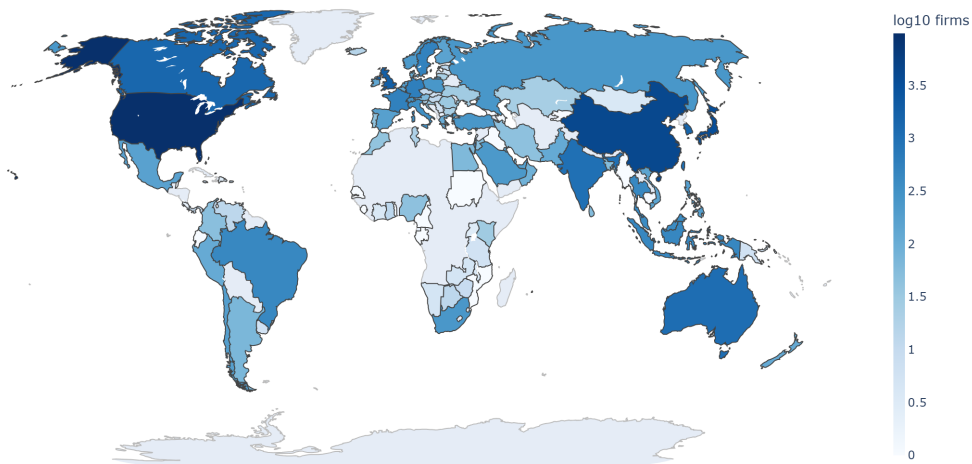


Figure 3: Source firms by country (logarithmic colour scale).

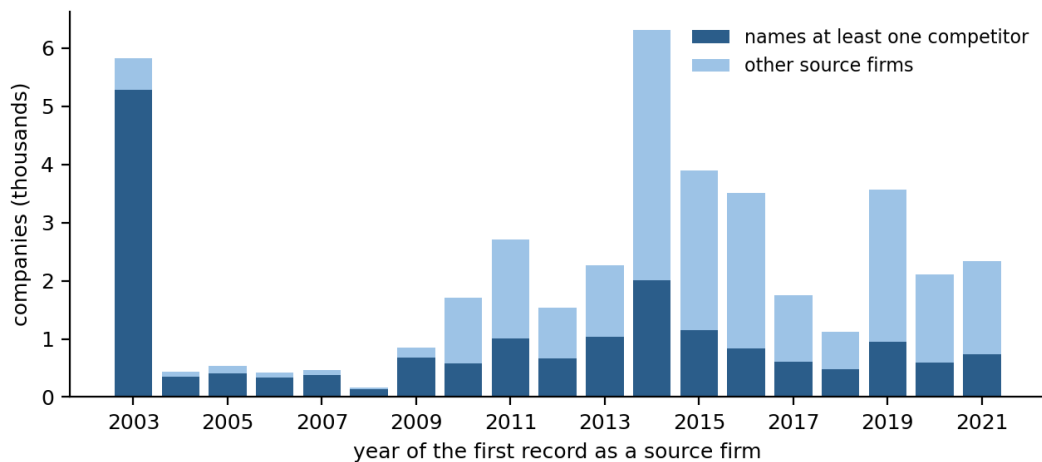


Figure 4: Source firms by the year of their first record, whether or not they name a competitor.

FactSet built its coverage region by region, adding the listed companies of North America from 2003, of Europe from 2011, of Asia from 2013, of the Pacific, the Middle East and Africa from 2014,

and of Latin America from 2016, and the steps of Figure 2 are these additions: the share of North America among the source firms falls from 89% in 2003 to 19% in 2021. These years are not first-record dates, however. Companies of every region appear from 2003, the foreign filers with the Securities and Exchange Commission first, since the analysts read their filings like those of a domestic company, and 1,850 of the 8,950 source firms outside North America that name a competitor have a competitor record before the year in which their region was added. We therefore date a firm’s coverage by its own first record as a source firm, whatever the type of that record, and we treat every earlier year as a year in which its relationships are unknown.

How relationships are recorded, and how we read them

Two features of the raw records matter for how we translate them into relationships: their direction and their dates.

Because each record is constructed from the disclosures of the firm whose documents identify the relationship, every FactSet record is directed. This direction, however, does not always characterize the underlying economic relationship. If A identifies B as a customer, the record runs from A to B , although the same relationship necessarily means that A is a supplier of B . A record running from B to A appears only if FactSet covers B as a source firm and B ’s own disclosures identify A as a supplier. This distinction matters because disclosure is asymmetric: a small supplier is much more likely to name a large customer than the large customer is to name that supplier, and the firms making the disclosures are public companies whereas the firms they name are more heterogeneous. We therefore treat supply-chain relationships and partnerships as bilateral economic links, using disclosures from either endpoint to recover them. Competitor records are different because their direction contains information: a firm’s designated rivals reflect its own characterization of its competitive set, and the firms that designate it are not necessarily firms it would itself name. We preserve this direction in the main analysis and examine undirected and mutual definitions as robustness alternatives, as Section 1 describes.

The two dates attached to a record are likewise dates of the record, not dates of the underlying relationship. FactSet’s user guide defines the start date as the date on which the record was first published and the end date as the date on which it was terminated, a blank end date marking a record that is still current, and it describes the closed records as past versions of the entry. A record therefore ends on the day it was retired in the database, and the data let us tell three reasons for that apart. Taking the competitor records, among the closed records, 21% are replaced by a new record of the same firm pair on the same day or the next. These cases are consistent with re-publication of the entry rather than interruption of the underlying relationship. A further 11% are followed by another record for the same firm pair after a gap, most often of one or two years, matching the rhythm of annual reporting and consistent with the rival disappearing from the recorded disclosures and later returning. The remaining 68% are the pair’s final observed record. In these cases the data alone do not distinguish between the source firm’s subsequent documents ceasing to name the rival and FactSet ceasing to maintain coverage of the source firm. The same broad pattern appears across the other relationship types. Over all records, 22% of pairs appear in multiple records of the same type; and, among the gaps between consecutive records, 77% are one month or less, most of them zero. Positive gaps have a median length of 396 days and cluster at whole-year intervals, as Figure 5 shows.

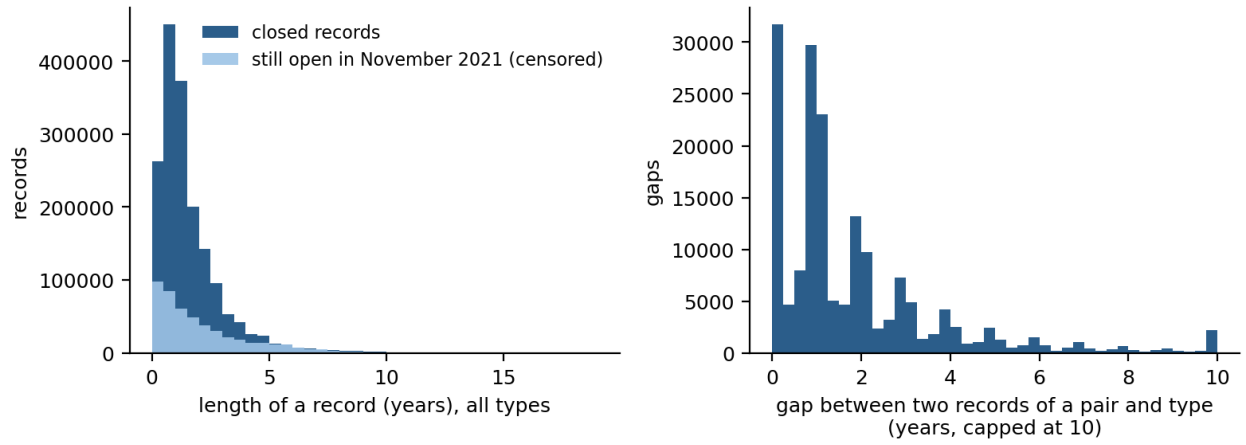


Figure 5: Left: the length of a record in years, all types; the records still open in November 2021 are censored at that date. Right: the gap between two consecutive records of the same directed pair and type, for the pairs with several records; the spikes at whole years seem to reflect the annual reporting cycle.

The dates themselves also bear the marks of database administration. They fall on working days and cluster on a small number of dates on which thousands of records of many different types were closed simultaneously; Figure 6 marks the largest of these dates: 12 July 2005 (16,004); 18 January 2013 (22,395); 6 February 2013 (8,339); 3 August 2017 (12,111). On the first, most affected records were re-opened the same day, a pattern consistent with a database migration. On the others, few were re-opened immediately, a pattern more consistent with a change or clean-up in coverage. Their concentration across many firms and relationship types makes it unlikely that these closure dates represent firm-specific corporate events, and together they account for 3% of all closures. Thus, neither observation record dates should be read as the date on which the underlying economic relationship itself began or ended.

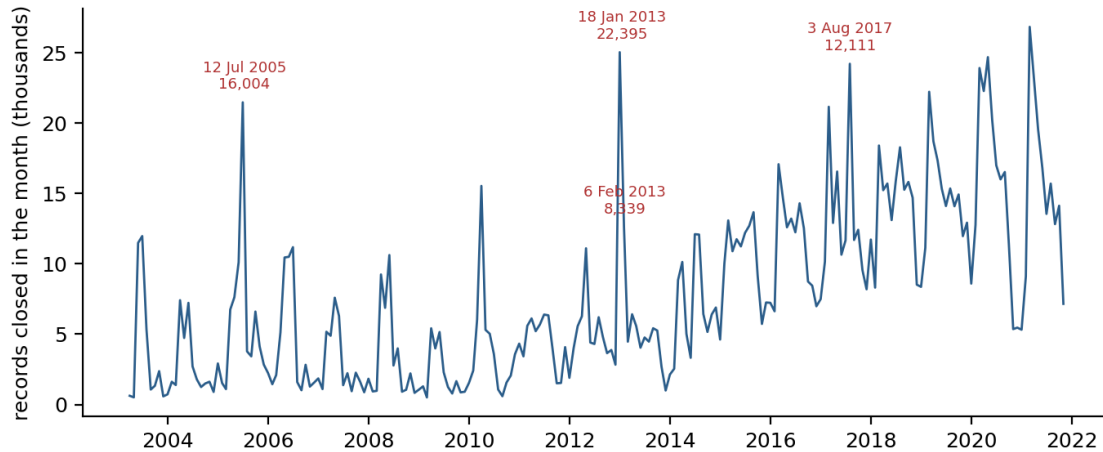


Figure 6: Records closed per month, all types, with the four largest single days marked.

We therefore read relationships at the calendar-year level and carry two principal definitions. Under the *calendar* reading, which is used in the main analysis, a pair is treated as related in every calendar year touched by any of its records. A deal involving a rival therefore counts for the focal firm in the fiscal period of its announcement whenever the two firms are related in the corresponding calendar year, even if the announcement date itself falls outside the exact span of the underlying record. This rule is insensitive to re-publication, to within-year gaps such as a record ending in June and its successor beginning in October, and to the batch closure dates. It does not, however, bridge calendar years with no record: 4.8% of pairs have at least one calendar year without a record between two years in which a

record is observed, yielding 72,731 such gap episodes in total.

The *gap-filled* reading addresses the subset of those episodes in which the source firm has no live record of that relationship type at all during the gap. Such episodes are the ones most consistent with an interruption in FactSet’s reading or maintenance of the source firm’s disclosures rather than the selective disappearance of that particular relationship. We therefore bridge only these gaps. This fills 10,405 episodes (14%) and adds 25,264 pair-years to the 4,919,257 pair-years obtained under the calendar reading. For competitor records, the rule fills 5,392 of 16,291 gap episodes, whose median length is 2 years, and increases the number of pair-years by 1.3%. By contrast, filling every gap between a pair’s first and last record would increase pair-years by 4.0%, but would also treat a rival that disappeared from a firm’s disclosures while the firm continued to name other rivals as if it had never disappeared. The data do not support that assumption. Figure 7 compares the three definitions. We also carry a fourth variant based on the exact record dates, under which a deal counts only if its announcement falls between the start and end dates of the raw record. For the reasons above, this stricter definition excludes some deals that the baseline annual readings treat as exposures.

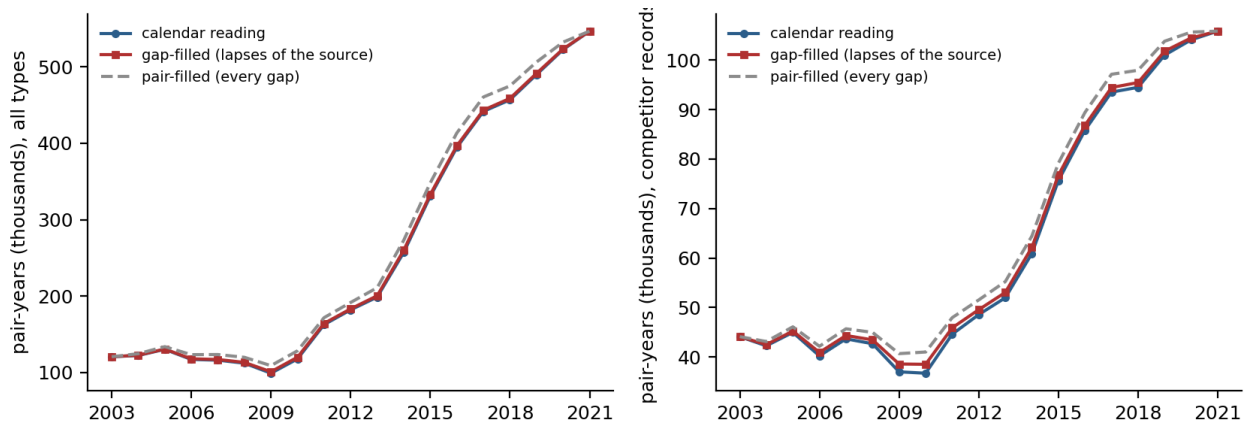


Figure 7: Pair-years per calendar year under the calendar reading, the gap-filled reading (a gap closed only inside a lapse of the source firm’s records of the same type) and the pair-filled reading (every gap closed). Left: all types; right: competitor records.

The rivalry network

Everything so far concerns the table as a whole. The paper works with the part of it that is about competition, and this subsection narrows the view step by step, from the records to the firms, from the firms to their rival sets, and from the sets to the network they form. Of the 2,209,235 records, 81% have at one end or the other a source firm that names at least one competitor, 70% as the source and 49% as the target, and the universe of the analysis is made of every entity that such a firm names or is named by, in any type of record, which comes to 212,674 of the 311,633 entities in the table. The competitor records themselves are 414,858 records over 305,850 directed pairs, written by 18,216 source firms in 104 countries about 75,902 named entities in 186 countries. A source firm’s rival set in a year is the set of entities it designates in a competitor record live in that year, and over 2003 to 2021 the calendar reading gives 1,198,305 rival pair-years. Table 2 follows these quantities year by year.

Table 2: The rivalry network by year: source firms with a live rival, named rivals, rival pair-years, and the median rival set.

year	source firms	named rivals	rival pair-years	median rivals
2003	4,701	10,377	44,124	8
2004	4,617	10,807	42,267	7
2005	4,632	12,029	44,984	8
2006	4,224	11,657	40,229	7
2007	4,393	12,561	43,677	8
2008	4,257	12,738	42,689	8
2009	4,449	12,312	37,012	6
2010	4,628	13,003	36,732	6
2011	5,019	15,739	44,615	6
2012	5,311	16,849	48,580	7
2013	5,634	17,919	51,981	7
2014	6,142	20,978	60,909	7
2015	7,174	25,463	75,644	8
2016	7,901	28,905	85,833	8
2017	8,530	30,867	93,542	8
2018	8,356	31,108	94,504	8
2019	8,856	32,383	101,014	8
2020	9,023	33,679	104,105	8
2021	9,173	34,900	105,864	8

The two sides of a competitor record are different populations, and this is the first thing to know about the network. The 18,216 firms that name competitors are listed companies whose documents the analysts read; 46% of them are American, and the eight largest countries are the United States (8,315), China (1,611), South Korea (907), Canada (827), the United Kingdom (696), Taiwan (554), Australia (453), Israel (363). The 75,902 entities they name need not be listed, need not be covered, and need not name anyone: 82% of them never appear as a source of a competitor record, and only 64% of the rival pair-years point to a rival that names competitors itself. The named entities are also far more concentrated than the naming firms. Over the whole window, 61% of them are named by a single firm, while 5,375 (7.1%) are named by ten firms or more, and the tenth of the entities that are named most accounts for 63% of all directed pairs. Figure 8 lists the twenty entities named by the most firms: Johnson & Johnson heads the list with 1,006 distinct firms naming it, and the list is made of the large pharmaceutical, technology and industrial groups that a firm in any of a dozen industries can plausibly call a competitor. These are the hubs of the network, and the reason a rival’s consolidation can touch many firms at once.¹

¹Three interactive companions to this subsection are available online at <https://salma-lemkhente.github.io/innovation-rival-consolidation/Appendix/interactive-graphs/>: a map of every entity of the rivalry network with headquarters coordinates, the source firms naming a competitor in one colour and the entities only named in another; a map of the rivalry flows between countries, in which every directed pair of firms incorporated in different countries counts once over the whole period and a slider keeps only the largest flows; and an industry-to-industry chord diagram of all the directed pairs, counted the same way by two-digit SIC industry, with the two directions between two industries combined and a slider that keeps only the largest chords. The firm-level map is drawn from the clean data of this appendix; the two others were drawn from the raw relationship file and show the same universe of firms.

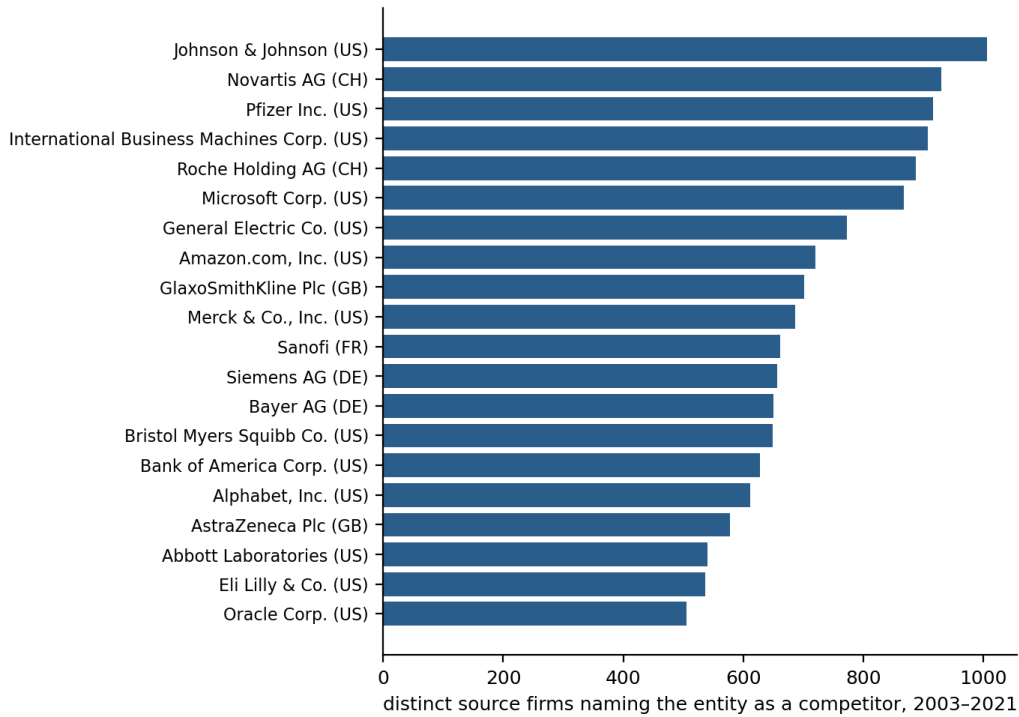


Figure 8: The twenty entities named as a competitor by the most source firms, 2003–2021, with the number of distinct firms naming each.

Seen from the naming side, the network is made of small, deliberate sets. Over the 117,020 firm-years of the window in which a source firm names competitors, the median set holds 8 rivals, the mean 10.2, and the ninetieth percentile 22 (Figure 9), and the median stays between six and eight across the nineteen years while the number of firms doubles, so that the network grows by adding firms rather than by thickening. The sets are also persistent. From one year to the next a firm keeps on average 90% of the rivals it named the year before, the whole of them in 57% of the firm-years, and on average 11% of a year's set is new; the years in which the coverage was extended show as small waves of renewal in Figure 10. A directed pair lasts a median of 3 years, and 15% of the pairs are seen in a single year only. This persistence is what makes a consolidation among a firm's rivals a well-defined event: the set that receives the shock is the set the firm has been naming for years, not a set that changes with every report.

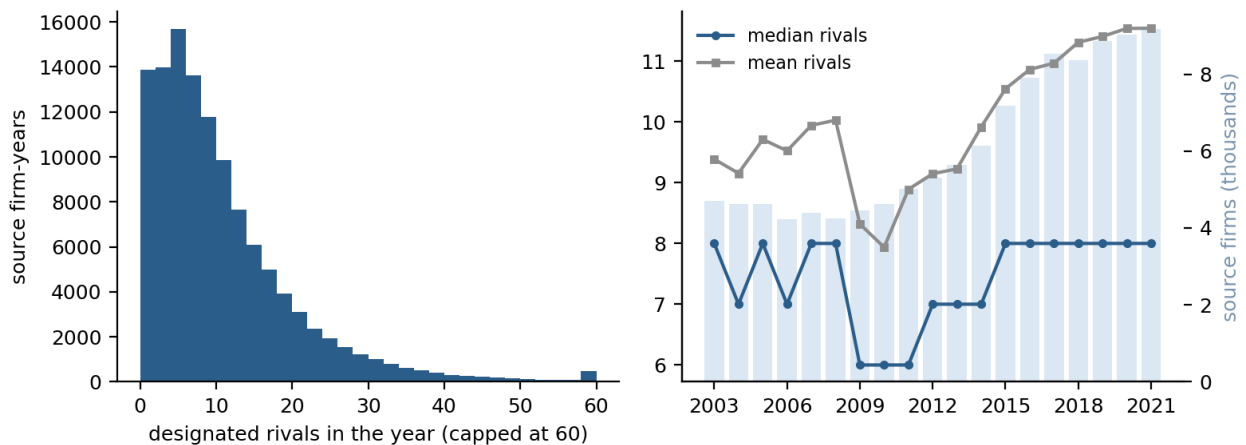


Figure 9: Left: designated rivals per source firm-year. Right: the median and mean number of rivals by year (lines) and the number of source firms naming competitors (bars).

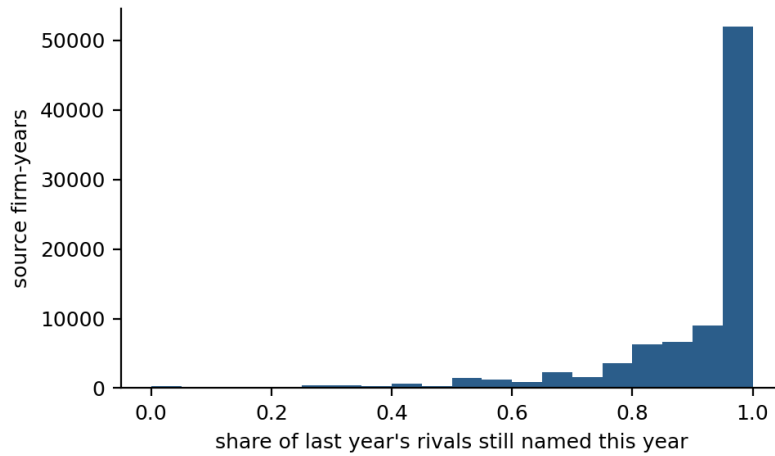


Figure 10: Year-to-year change of a firm's rival set: the share of last year's rivals still named this year, per source firm-year.

Naming and being named are related but distinct. Figure 11 shows, on the left, how many firms name an entity in a year, a distribution in which 63% of the entity-years are a single firm and the most named entity is named by 509, and on the right the share of a firm's designated rivals that name it back, which is zero for most firm-years. Only 13.0% of the directed pairs exist in both directions, 10.8% of the rival pair-years point to a rival that names the firm back, and 46% of the firm-years have at least one such mutual rival. Among the 14,003 firms that both name and are named, the two counts are correlated but loosely (Figure 12), and 21% of them are named by more firms than they name. Asymmetry is the rule rather than the exception, which is why the direction of the records is kept and why the firms that name a firm are treated as a different set from the firms it names.

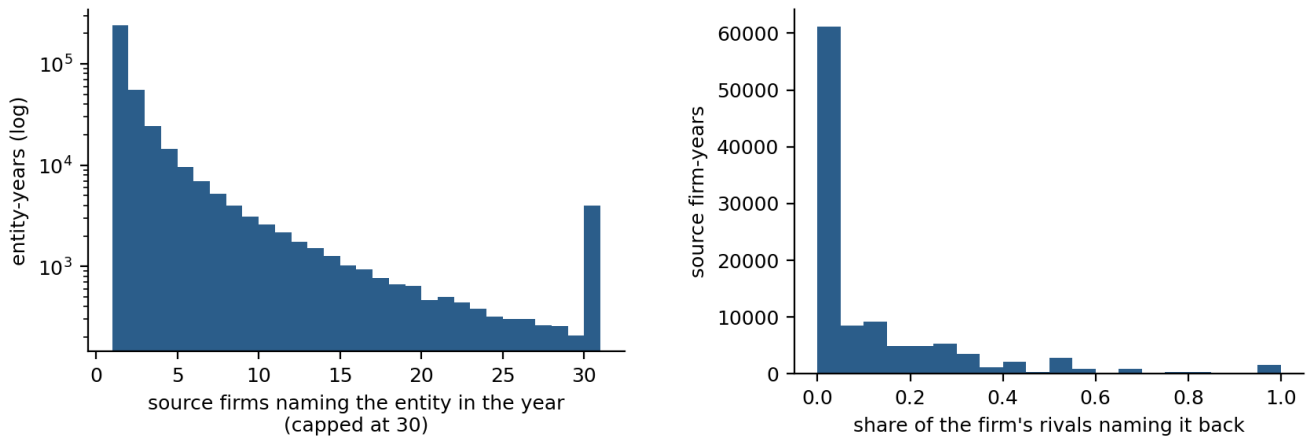


Figure 11: Left: how many source firms name an entity in a year (logarithmic count). Right: the share of a firm's designated rivals that name it back, per firm-year.

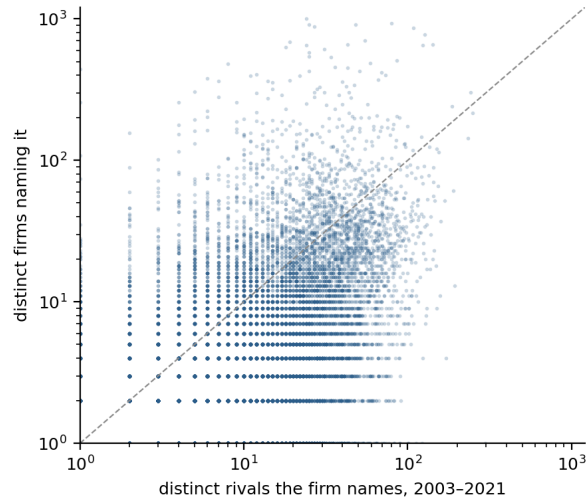


Figure 12: For the firms that both name and are named, the number of distinct rivals the firm names against the number of distinct firms naming it, 2003–2021, one dot per firm on logarithmic scales; the dashed line is equality.

Taken together, the pairs form one connected network rather than a collection of separate markets. Of the 80,115 entities that name or are named, 98.5% belong to the largest connected component when the direction of the pairs is ignored, and the remaining 315 components are small. The network is wide rather than tight: two entities of the main component are a median of 5 steps apart along rival pairs, only 26% of the pairs of entities are within four steps of each other, and the share of a firm’s rivals that are rivals of one another, the clustering of the network, is 0.16. This is the structure the rivals-of-rivals set exploits, and it is also why a consolidation two steps away is a genuinely different exposure from a consolidation among the firm’s own rivals.

Rivalry crosses borders and industries. Over the rival pair-years, 54% join two firms of the same country and 57% join two firms of the same two-digit SIC industry, where both codes are known, which is the case for 97% of the pair-years; the same-country share falls from two thirds in 2003 to one half in 2021 as the coverage spreads beyond North America (Figure 13). The headquarters of a firm and of its rival lie a median of 2,170 kilometres apart, and 23% of the pairs lie within 500 kilometres of each other (Figure 13). A rival set defined by the firm’s own designations is therefore a different object from an industry peer set: it reaches into other industries and other countries, and it leaves out most of the firms that share the firm’s industry code. Finally, the competitor records behave as the records of every type do: the median record lasts 1.6 years, 21% of the directed pairs appear in several records, and 71% of the gaps between two records of a pair are of a month or less (Figure 14). Table 3 gathers these quantities in one place.

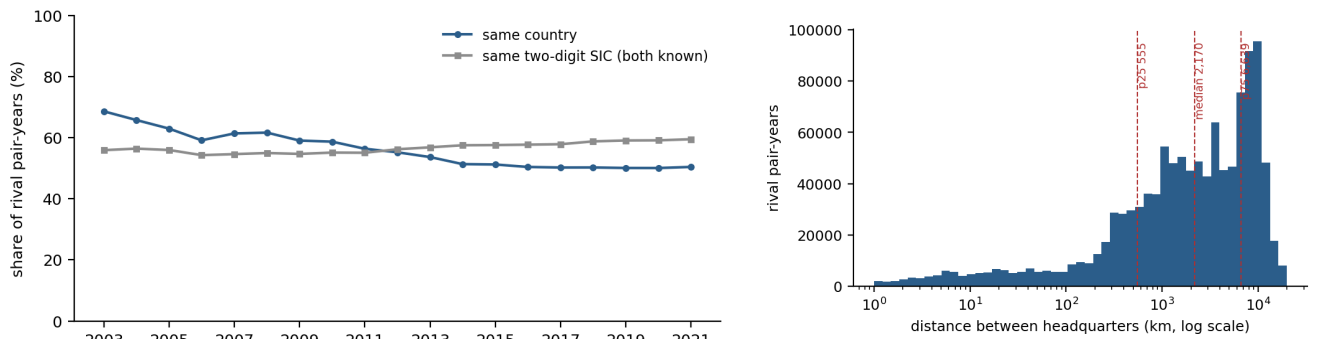


Figure 13: Left: the share of rival pair-years within one country and within one two-digit SIC industry, by year. Right: the distance between the headquarters of the two firms of a pair, for the pairs with coordinates.

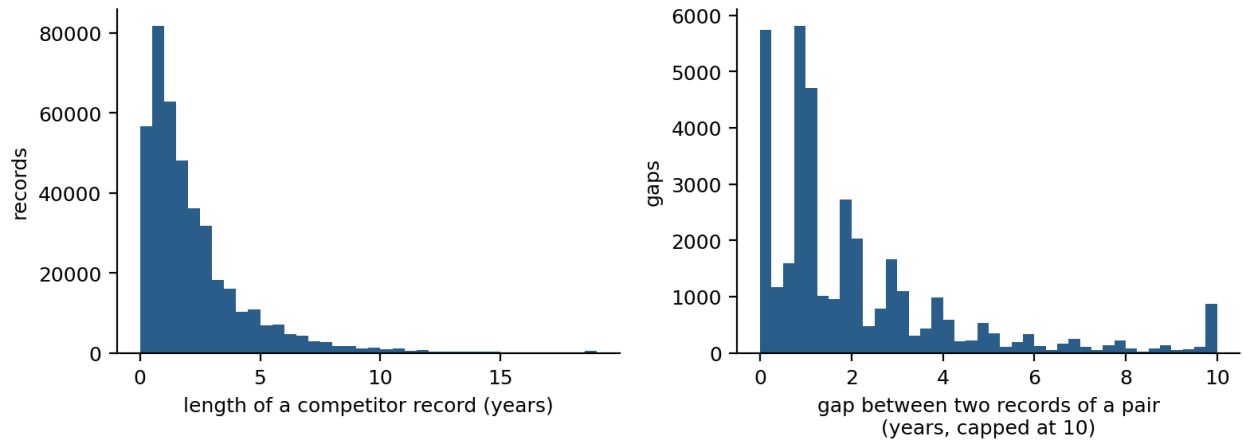


Figure 14: Left: the length of a competitor record in years. Right: the gap between two consecutive records of the same directed pair, in years, for the pairs with several records.

Table 3: The rivalry network in numbers, 2003–2021.

Source firms naming at least one competitor	18,216
Named rivals (entities designated by a source firm)	75,902
Named rivals that are themselves source firms naming competitors	14,003
Competitor records (raw, after successor mapping)	414,858
Directed rival pairs, 2003–2021	305,850
Rival pair-years, 2003–2021 (calendar rule)	1,198,305
Designated rivals per source firm-year: median / mean / p90	8 / 10.2 / 22
Times an entity is named per year: median / max	1 / 509
Reciprocity of the directed pairs (both directions exist)	13.0%
Rival pair-years whose rival names the firm back	10.8%
Rival pair-years within one country / within one two-digit SIC	54% / 57%
Median distance between headquarters (km)	2,170
Largest connected component (share of firms in the network)	98.5%
Source firm-years in the window without a live rival record	8,275 (7.1%)

Beyond the designated set

The designated rivals are the firm’s own reading of its competition, and the main analysis uses them. Because the records are directed and the network is known around every firm, three other sets can be built from the same records, and the analysis asks whether a rival’s consolidation reaches the firm through them as well; Figure 15 shows their sizes. The *undirected set* adds to the firm’s designated rivals the firms that designate it without being designated back, which we call its namers. Where a firm has namers it has a median of 3 of them, so that the undirected set is larger than the designated set in 59% of the firm-years and holds a median of 11 firms; a consolidation among the namers is one that the firm’s own documents do not point to, but that reaches it through firms that see it as a competitor. The *mutual set* keeps only the rivals that name the firm back in the same year. It exists in 47% of the firm-years and holds a median of 2 firms where it exists, and on average 12% of a firm’s designated rivals are mutual; it is the strictest reading of rivalry, the pairs that both sides acknowledge. The *rivals of rivals*, finally, are the designated rivals of the firm’s rivals that are source firms themselves, without the firm and without its own rivals. They are known in 94% of the firm-years, whenever at least one rival names competitors, and they number a median of 36 where they are known; a consolidation among them is a consolidation two steps away, and it is the channel through which a shock can reach a firm that neither names nor is named by the consolidating firms.

The suppliers, customers and partners of a firm are built in the same way from the other record

types, reading each record from both of its sides, and they are far more common than the alternative rival sets: in 88% of the firm-years the firm has at least one live supplier or customer and in 81% at least one partner (Figure 16). The estimation uses these sets as the channels through which a consolidation can reach a firm other than through its designated rivals, and as the controls that separate the rival channel from the rest, and Section 4 describes how each of them enters.

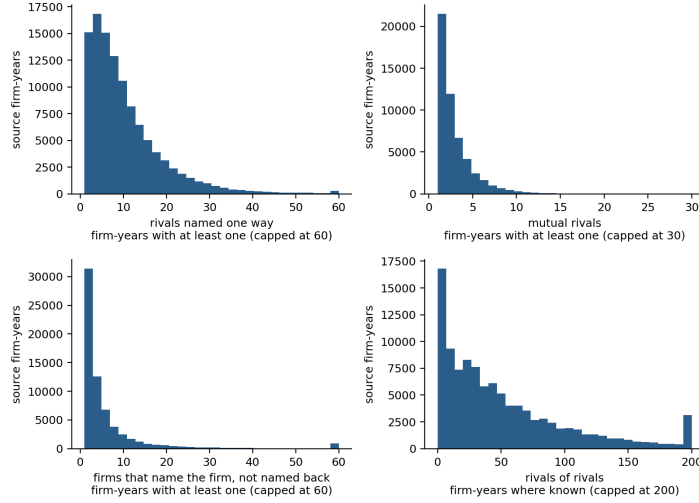


Figure 15: The four rival sets per source firm-year, calendar reading: the designated rivals; the undirected set; the mutual rivals, over the firm-years with at least one; the rivals of rivals, over the firm-years in which at least one rival names competitors.

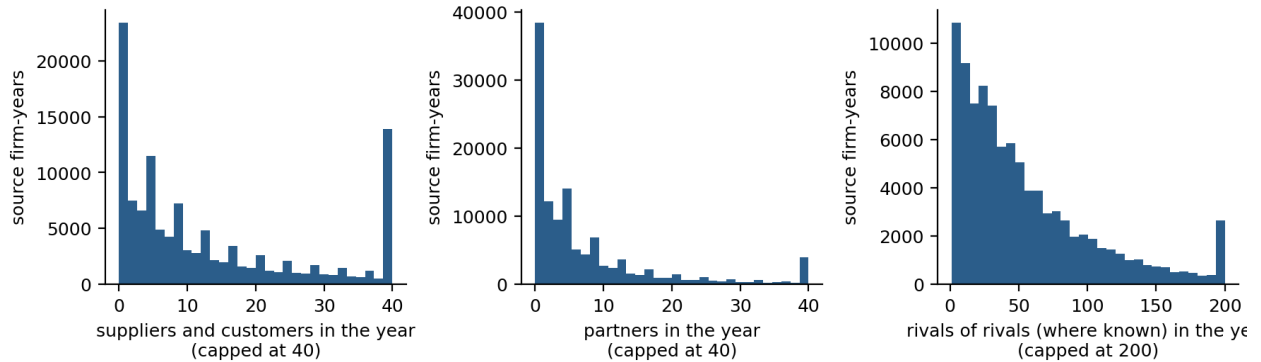


Figure 16: Suppliers and customers, partners, and rivals of rivals per source firm-year.

2 The financial panel

The source firms of Section 1 that name at least one competitor are the firms whose innovation the paper studies, and we call them the *focal firms* from here on: there are 18,216 of them. This section builds their financial panel. The panel has two uses. It supplies the main outcome of the paper, the firm's expenditure on research and development, and it supplies the characteristics on which the firm is conditioned before a shock reaches its rivals: its size, its sales, its cash, its investment, and the accounting conventions under which it reports. Neither is available from FactSet's relationship data, which know a firm only as a node of the network. The accounts have to be fetched from a vendor of financial statements, and the vendor identifies companies in its own way. The section explains how each focal firm was matched, year by year, to the accounts that are its own, why that match is harder than it looks, what was done for the years in which the first vendor has no accounts, and how the outcome variable was harmonised across sources. It ends with the panel as the estimation reads it.

What has to be linked, and why an identifier does not settle it

The accounts come first from S&P's Compustat, which reports the financial statements of listed companies of North America and of the rest of the world in two sets of tables (S&P Global Market Intelligence 2026), and identifies a company by a permanent key, the **gvkey**. What the panel needs is a yearly crosswalk: for every focal firm and every fiscal year that touches its span, the **gvkey** whose accounts are the accounts of that firm in that year. The span is the window of the firm, the calendar years between its first and its last competitor record, extended backwards by two years so that the controls of the first window year can be lagged; the two earlier years are flagged and never used to measure exposure.

No such crosswalk exists. Neither FactSet nor S&P publishes a table from the FactSet entity identifier to the **gvkey**, and the link has to be assembled from identifiers that the two vendors happen to share, the CUSIPs and ISINs of securities and the registrant number of the Securities and Exchange Commission. These identify securities and registrants, not the two objects that have to be matched, and the difference is where the difficulty begins. FactSet identifies entities, and its documentation is explicit that an entity is any organisation it tracks, a listed company as much as a subsidiary, a holding company or a company that no longer exists; an entity carries no accounts of its own. Compustat identifies a company by the continuous series of financial statements it reports under one **gvkey**. For a listed company that is never reorganised the two coincide. They part company at exactly the events this paper studies, because the two vendors follow opposite conventions at a reorganisation, each stated in its documentation. Compustat follows the accounting: at a merger it keeps the acquiring company's record alive and lets it absorb the combined accounts, and in a reverse acquisition the record of the accounting acquirer survives under the other company's name. FactSet follows the organisation: when a transaction creates a new company it opens a new entity for it and makes the old one extinct, often renamed with the suffix "/Old/", and when a holding company is placed above an operating company it opens an entity for the holding and reclassifies the operating company as its subsidiary. When two companies merge, then, Compustat keeps one record and FactSet opens a new entity, and a security identifier that linked the entity and the **gvkey** correctly before the event links them wrongly, or links the wrong pair, after it.

The Dow Chemical Company shows every difficulty at once. On 31 August 2017 Dow and DuPont combined under a new parent, DowDuPont, which separated in 2019 into three listed companies, one of which, the renamed DuPont de Nemours, kept the parent's shell. FactSet records the episode as a succession of entities: The Dow Chemical Co. is one entity throughout and names rivals from 2003 to the end of the data, a second entity was opened for DowDuPont and now bears the name DuPont de Nemours, and two more were opened in 2019 for the new Dow, Inc. and for Corteva. Compustat records it as the life of one record: Dow's **gvkey** survived the merger of equals, carried DowDuPont's consolidated accounts for 2017 and 2018, and carries DuPont de Nemours's accounts today, under that name; Dow's own business continues under a new record that Compustat opened in 2019. For the single entity The Dow Chemical Co. the answer to "which record carries its accounts" is therefore a sequence: its own record until 2016, its parent's record in 2017 and 2018, and no record at all from 2019, since the record that carries the Dow business after the separation is connected by every identifier to the entity Dow, Inc. and to no other. A link between an entity and a record can be right for some years and wrong for others; names mislead in both directions, since the right link joins "The Dow Chemical Co." to a record now called "DuPont de Nemours" while the wrong pairing of FactSet's DuPont entity with the pre-2017 accounts shows a perfect match; and whether a parent's accounts may stand in for a firm's own is a decision that has to be taken explicitly and recorded.

How the crosswalk was built

Four requirements follow, and they organise the construction. A link must carry dates and be built from dated evidence, so that it can hold for some years and not for others. Names can corroborate a link but cannot establish one, and since neither vendor dates its names, FactSet giving up to six kinds of

name per entity without a date and Compustat’s principal name being the one current at the day of the download, names are compared as pools of every name each side ever carried. Where dated evidence does not settle a link, what is needed is a judgement on the lineage of the company, year by year, with its corporate history in hand. And the cases in which a firm has several records, or a record several firms, cannot be settled by a rule of precedence, because the same configuration is right in one corporate history and wrong in another.

The candidate links are assembled from a chain of dated tables: FactSet’s dated association of an entity with its securities and of a security with its CUSIPs and ISINs (FactSet Research Systems 2021*a*), then Compustat’s dated identifier history, or CRSP’s dated CUSIPs and its own dated link to Compustat (Center for Research in Security Prices, LLC 2026), for the securities listed in the United States; the SEC registrant number, which neither vendor dates, and a link table that WRDS infers partly from ticker symbols (Wharton Research Data Services 2026*a*) are kept as evidence and never count as dated. The chain proposes 20,731 candidate pairs for 17,921 focal firms, 98.4% of them, each pair with the days on which every step of its chain holds. A pair is accepted without further examination, and called cleared, only when two independent kinds of evidence agree: at least one route links it through steps that all carry a vendor’s dates, on days that overlap the firm’s span, and at least one name that FactSet gives the entity matches, after normalisation, at least one name that Compustat or CRSP ever gave the record. Both tests are needed. A dated link without a name match is a good link seven times in ten, the company having been renamed in a way that no name source records, but three times in ten it is a trust that repackaged the company’s debentures and shares its CUSIP on exactly the same days, or a parent, or a namesake; a name match without a dated link is the wrong generation of a company that divided and passed its name on. A cleared pair holds only on the days its dated links cover: the fiscal years of the span that those days do not reach, typically the two years before a listing or the years after a reorganisation, are not filled by rule but checked by the review described next, for those years alone.

Everything the rule does not settle is judged on the corporate history of the two companies, through a combination of web searches and agentic reviews with a large language model, set up to confirm or reject each match. The question is the one this section began with: when one reads the accounts filed under this record in year t , are they the accounts of the company that this entity denotes in year t ? Answering it means establishing that the two denote the same company at the time, which their names today cannot show, since a name may have passed to another company and a company may have changed its name. The review therefore reconstructs the lineage of the two, through their renamings, mergers, divisions and changes of listing, and judges whether the match is plausible in each year. Its answer is a relation, the same company, a subsidiary that the entity controls and that is substantially all of it, a parent whose main business the entity is, an affiliate, an unrelated company, or unsure, together with the years in which it holds and the lineage that supports it. The acceptance is decided afterwards in code, from the relations, so that the policy is stated once and applied uniformly: the same company is accepted for the years given; a subsidiary’s record is accepted for its parent only when it is substantially all of the parent; a parent’s record is accepted for a subsidiary only when the subsidiary is the parent’s main business, and such links are flagged so that the estimation can run without them; affiliates, unrelated companies and unsure verdicts are never accepted; and a link counts only in fiscal periods for which Compustat reports total assets or sales. Every verdict is kept with the code, with its reasoning, and a random sample of the decisions of every kind was checked by hand against the companies’ histories.

The accepted pairs, turned into link-years, can still contradict one another, a firm having two records in one fiscal year or a record two firms, and these contradictions are the trace that FactSet’s convention of opening a new entity leaves when it meets Compustat’s convention of keeping the record: an entity marked “/Old/” and its successor both linked to the one record, a parent and its subsidiary linked to the parent’s accounts, or the year in which a record changes hands. Each cluster of colliding links was examined as a whole, with every firm and every record involved in view, under a stated order of preference, the firm’s own record before a parent’s or a subsidiary’s, and the record whose accounts

are the company’s own before a wider group’s or a converted copy. A record may remain attached to two firms in a year only for one of two named reasons, that FactSet holds two entities for what is one company in that year, or that a parent and its subsidiary have only the parent’s consolidated accounts; such firms receive a common group identifier and are treated as one reporting unit in the years they share. Dow illustrates the outcome: The Dow Chemical Co. has its own record by rule for the fiscal years to 2017, its parent’s record for 2018 with the parent flag, and nothing from 2019; the entity DuPont de Nemours has the same record from 2017, sharing 2017 and 2018 with the former as parent and subsidiary; the entity Dow, Inc. has the new record from 2019 by rule and, on review, for the two earlier years in which that record carries the accounts of its accounting predecessor.

The crosswalk has 138,662 firm-years for 17,655 of the 18,216 focal firms (96.9%) and 17,523 records; 104,410 of the firm-years reach into the firms’ windows, for 17,053 firms (93.6%). The rule supplies 91% of the firm-years and the review 9%; 12,656 firms owe every one of their years to the rule, and for most of the 4,999 others the reviewed years are only the years before the window. The links that join a firm to a record other than its own are few, 110 parent firm-years and 8 subsidiary firm-years, and flagged. A firm keeps one record throughout in all but 169 cases, and in 1,126 firm-years the record is shared with another focal firm, by 437 firms forming 214 reporting units: 999 firm-years because FactSet holds two entities for one company and 127 because a parent and its subsidiary have only the parent’s accounts. Table 4 gives the levels.

Table 4: How the accounts of each firm-year of the panel were linked to the firm.

how the accounts of the firm-year were linked	firm-years	of which in the window
the firm’s own Compustat record, cleared by rule	126,398	100,306
the firm’s own Compustat record, accepted on review	12,146	4,010
a parent’s record, the firm being its main business	110	86
a subsidiary’s record, being substantially all of the firm	8	8
the buyer’s consolidated accounts, after the firm was acquired	1,597	1,597
FactSet Fundamentals, the firm’s own series	2,384	1,168
all	142,643	107,175

A crosswalk of this kind is judged as much by what it leaves empty as by what it fills. Of the span years, 15,141 remain without a Compustat record, 12,755 of them inside the windows, for 5,996 firms, and every one has a recorded reason. The first reason, by far, is not a failure of the match: in 11,495 of the empty window years the firm has an accepted record that reports no accounts for that year, because the company had been acquired, taken private or delisted while FactSet continued to record it as naming rivals, or because, as with Dow after 2018, the business continues under a record that every identifier attaches to another entity. In 946 no route proposes any record at all, half of these firms having no security identifier in FactSet; the rest are years in which the link does not hold, years removed when colliding links were resolved, and firms for which no candidate was confirmed.

Two more sources of accounts, and what they add

FactSet itself publishes annual accounts, in its Fundamentals database (FactSet Research Systems 2021*b*), keyed on its own securities, so that they can be reached without any crosswalk between vendors; and Bureau van Dijk’s Orbis (Bureau van Dijk 2026) collects the filed accounts of companies whether or not they are listed. We nevertheless take Compustat first and the others only for the years it leaves without accounts, and the reason lies in the variable we care most about. A firm-year can show positive research and development, a reported zero, or no value, and the last two are different states of the world, since a firm that reports no R&D line may well do research. In Compustat the distinction is stable. In FactSet it is not, and the instability follows the calendar rather than the firms: where Compustat has no value, FactSet reports a zero in one firm-year in ten in 2001, in one in three in 2010 and in one in twenty-five in

2018, and thousands of firms that never show positive R&D alternate in FactSet between years of zero and years without a value. A panel built on FactSet’s item would move firms between “no R&D” and “unknown” by vintage of collection, most of all around 2009 to 2013, which is inside our period. FactSet also warns that the history of a security can be deleted or stitched together with another company’s at a merger, which is the event we study.

A row of the panel therefore takes all its variables from one vendor, so that a ratio is never formed from two vendors’ definitions. FactSet’s accounts are reached through its dated table of the securities each entity held, and from the primary issue only, since an entity often reaches several series, the home listing and one or more depositary receipts, and FactSet re-points a receipt from one entity to another with its whole history: without the rule the entity DuPont de Nemours would show, for 2000 to 2016, Dow’s sales in Brazilian reais. Each variable pairs a Compustat item with the FactSet item that measures the same thing, and the pairing was tested rather than assumed on the 126,165 firm-years that both vendors report for the entity’s own record with a twelve-month period: in dollars, total assets are within five per cent of each other in 96.3% of them and sales in 94.2%, employees in 92.0%, and R&D in 82.2% (Figure 17). The test changed one pairing, FactSet’s total capital expenditure being replaced by its expenditure on fixed assets, and it retired one variable, operating income before depreciation, which the vendors draw differently and which is kept for Compustat rows only. Money is converted to United States dollars with one table of monthly exchange rates for every vendor, at the month-end rate of the period end for balance-sheet items and at the average rate of the period for flows; the accounting standard of each set of accounts is carried in four classes (58% of the firm-years are under US GAAP, 32% under IFRS, 10% under a local standard); and fiscal periods that are not twelve months long, which arise when a company changes its year-end, are kept with their length and flagged rather than annualised (517 firm-years).

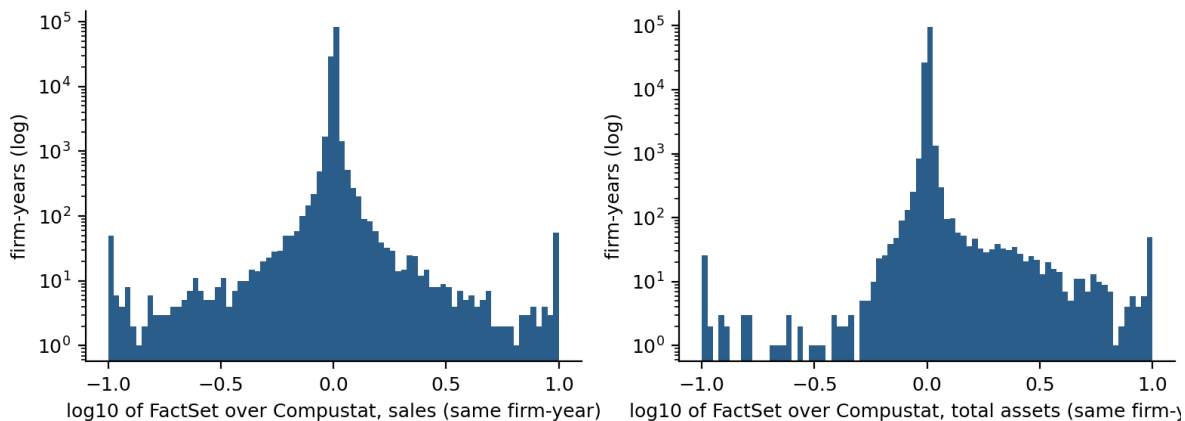


Figure 17: FactSet Fundamentals against Compustat for the same firm and fiscal year, the firm’s own record and a twelve-month period: the logarithm of the ratio of the two vendors’ sales (left) and total assets (right), in dollars.

The comparison of the two vendors is also an independent test of the crosswalk, since FactSet’s accounts were never used to accept a link. If a record were another company than the entity, its sales would differ from FactSet’s in most years, by a wide margin, and always in the same direction. The firms that show that pattern were examined one by one against the companies’ filings, and none is a wrong link: the two vendors measure revenue differently for casino operators, banks and payment processors, or one vendor’s series is faulty, which total assets, identical to the decimal, confirm. The same question was put to every firm for which FactSet fills a year, and the filled years that turned out to be another company’s, the first consolidated year of a merged group carried by the entity marked “/Old/”, the statutory accounts of a legal shell after a takeover with sales of zero where the business had billions, were removed, each with its reason, in a file kept with the code. Orbis was examined in the same way for the years that neither vendor covers: its companies were reached by every ISIN FactSet ever gave the entity

and by the entity's LEI, cleared only with a name match, its items were calibrated against the panel on the firm-years they share before any was used, and its filled years were checked like FactSet's. It fills 825 firm-years of 566 firms, most of them years before a listing; these rows enter the panel converted to dollars as the others are, consolidated accounts only, with their source recorded so that a ratio is never formed across vendors.

One more source of accounts is the firm's buyer. Of the focal firms bought in a completed control deal, FactSet keeps recording a third naming rivals a year or more after the deal, so that the firm remains a disclosing unit with its own rival set while its accounts are consolidated in the buyer's. For the years after the deal in which the firm still names rivals and has no accounts of its own, its row carries the buyer's consolidated accounts under a level of its own, 1,597 firm-years for 682 firms; the estimation chooses between the firm's own accounts and this wider sample by the level, and the main analysis uses the firm's own.

The panel

The panel has 142,643 firm-years for 18,006 focal firms (98.8%): 140,259 rows from Compustat, including the 1,597 buyer-consolidated ones, and 2,384 from FactSet Fundamentals, for 1,037 firms of which 343 have no Compustat year at all. Each row gives the firm, the fiscal year and its period end; its source and the vendor's identifier of the series, so that every value can be traced to a vendor's record; how the link was accepted; the currency in which the accounts were reported, the accounting standard and the length of the period; whether the period precedes the firm's window; the accounts, in millions of dollars, and employees; and R&D with its status. Figure 18 follows the window firm-years year by year and shows where their accounts come from: of the 116,898 window firm-years, 91.1% have accounts, 88.6% from the firm's own Compustat record (85.2% cleared by rule, 3.4% accepted on review), 0.1% from a parent's or a subsidiary's record, 1.4% from the buyer's consolidated accounts and 1.0% from FactSet, while 8.9% have none. The share without accounts is lowest in the middle of the period and highest at the two ends, at the start because the coverage of the relationship data begins before the accounts of many small firms, and at the end because a firm that was bought or delisted goes on naming rivals for a while. Table 5 describes the variables over the window firm-years with the firm's own accounts: the median firm has sales of 417 million dollars, assets of 791 million and 1,459 employees, and 55% of these firm-years belong to American firms, a share above the firm count of Section 1 because American firms are covered longest and have accounts most often.

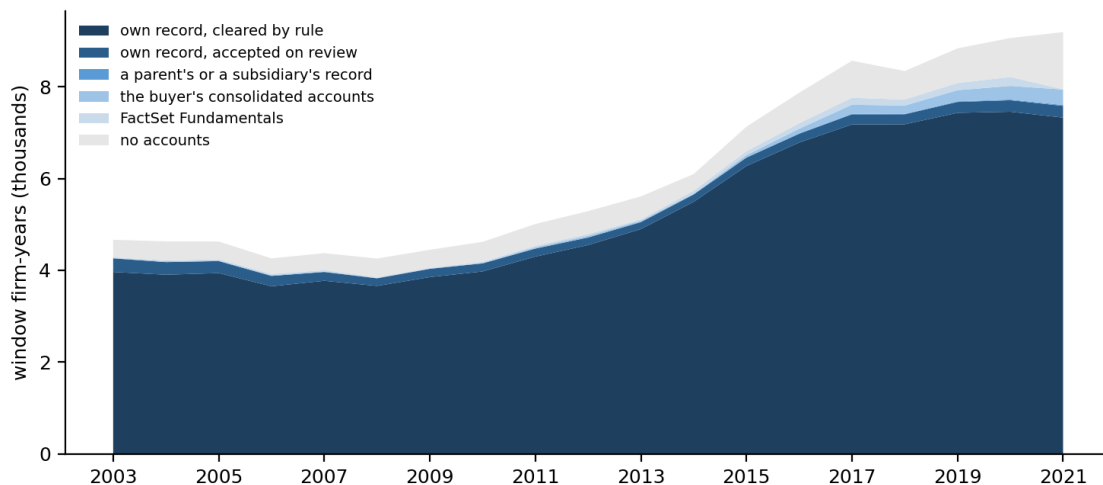


Figure 18: Window firm-years by fiscal year and the source of their accounts.

Table 5: The variables of the financial panel over the window firm-years with the firm’s own accounts, in millions of US dollars.

	firm-years	p10	median	mean	p90
Sales, million USD	101,006	12	417	4,168	7,817
Total assets, million USD	103,584	41	791	13,857	14,730
Employees	89,361	61	1,459	12,227	26,906
R&D, million USD (positive firm-years)	50,803	1.2	17.4	177.1	236.0
R&D over sales, % (positive firm-years, capped at 100)	48,897	0.4	6.2	18.7	80.7
Capital expenditure over assets, %	99,631	0.1	2.4	4.1	9.6
Cash over assets, %	101,027	1.8	12.3	21.4	58.1

Research and development

The outcome variable needs three decisions, and the section ends with them. The first concerns what R&D is. Until 2008 a United States acquirer expensed at once the part of a purchase price that it attributed to the target’s unfinished research; Compustat includes this charge in its R&D item and reports it separately, and FactSet’s item excludes it by definition, which we verified on the firm-years that carry the charge in both vendors. The charge is not research effort but part of the price of an acquisition, and in a study of mergers it would raise the acquirer’s R&D mechanically in the year it acquires. The measure of the paper is therefore R&D net of acquired in-process R&D: Compustat’s item less the charge, in the 2,539 firm-years that carry one, floored at zero in 14 of them, and FactSet’s item as it is; the raw figure is kept for comparison.

The second concerns zeros. A Compustat zero is a zero: 7,694 firm-years report one, and in 968 more the item is empty with Compustat’s code for an insignificant figure, which we record as a zero. A FactSet zero, for the reason given above, is accepted only when Compustat shows that the entity never reports positive R&D or does not cover it (161 firm-years) and is otherwise set to missing (40). A missing value is never turned into a zero, and the estimation carries the missing values as such, with an indicator, in the variants that impose a zero. The rule of one source per firm-year leaves unused the firm-years in which Compustat has no R&D value and FactSet a positive one for the same entity and period; a second measure takes FactSet’s value in 1,651 such firm-years where the two vendors otherwise concur on that very firm-year, on the standard, the period length, assets and sales, and it is used for robustness only, because these firm-years cluster in 2012 and 2013, the signature of a change in FactSet’s collection rather than in firms’ research.

The third is what the panel then shows, in Figure 19 and Figure 20. Over the 103,602 window firm-years with the firm’s own accounts, 49.0% report positive R&D, 6.5% a zero, and 44.5% no R&D line; 8,728 firms report positive R&D in at least one window year and 8,274 never do. The positive share moves little over the period. Among the positive firm-years the median R&D is 17.4 million dollars and the ninetieth percentile 236 million; R&D over sales has a median of 6.2% and a ninetieth percentile of 81%. The estimation takes the logarithm of R&D over the positive firm-years as its main outcome, the logarithm of one plus R&D and the inverse hyperbolic sine over the firm-years with a reported zero as well, and R&D over lagged assets as an intensity, each in a version that keeps the missing values missing and, for the transformations that admit it, in a version that imposes a zero with the indicator; Section 4 states the samples.

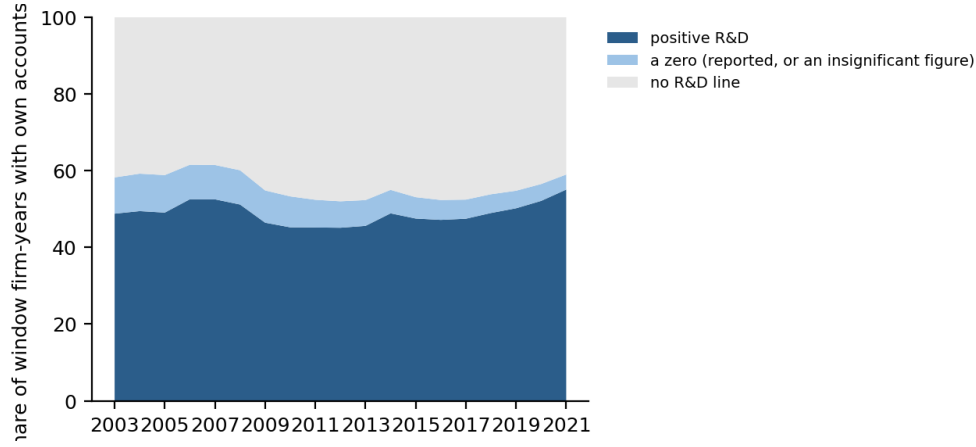


Figure 19: Research and development over the window firm-years with the firm’s own accounts, by fiscal year: positive, a zero, or no R&D line.

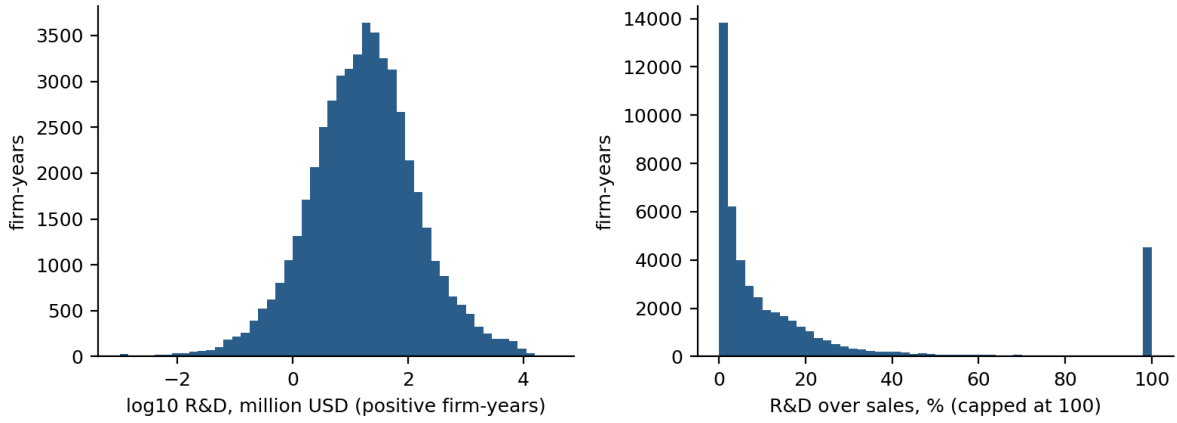


Figure 20: Left: R&D net of acquired in-process R&D over the positive window firm-years, in millions of dollars, logarithmic scale. Right: R&D over sales for the same firm-years, capped at 100 per cent.

3 The patent panel

The second outcome of the paper is a firm’s patenting: the patents it applies for in a year and the inventions, counted as patent families, that it originates. Patents have no firm identifier. A patent carries the name of its assignee as the applicant wrote it, and that name raises three questions that an identifier would have settled. Whose name is it: “Siemens Aktiengesellschaft”, “Siemens Medical Solutions USA”, “Osram Opto Semiconductors” and “BSH Bosch und Siemens Hausgeräte” all appear on patents that published datasets attribute to Siemens, and they are, respectively, the firm, a subsidiary, a subsidiary for some years, and a joint venture. When was it that firm’s name, since companies are renamed, acquired and divested, and a name that belongs to a firm in 2005 may belong to another in 2015. And which of our firms is meant, since Section 2 showed that FactSet can hold two entities for what Compustat records as one company. This section explains how the names were validated and dated, how the patents of all offices were then attached to the firms through them, and what the panel contains.

Why the published bridges are candidates and not answers

Several public datasets link United States patents to listed firms, and the literature uses them as given: the links of Kogan et al. (2017) to CRSP’s security identifier, those of Ma (2020, 2025) and of Dyèvre and Seager (2023) to Compustat’s `gvkey`, the firm identifiers of Arora, Belenzon and Sheer (2021) with

their dated subsidiary names, in the release of Arora et al. (2024), and a scored link published by WRDS (Wharton Research Data Services 2026c). Through the crosswalk of Section 2 they reach several million patents of our firms. We use them as a source of candidates and validate every candidate, for two reasons. They date ownership differently, by the application year, the filing year or the grant, so that around a merger they legitimately disagree; and they contain large errors that their agreement does not reveal, because they are not independent of one another: four of the five attribute the patents printed “Robert Bosch GmbH” to Parker-Hannifin, and others place Matsushita’s patents under Sumitomo Electric or Fuji Photo Film’s under Fujitsu. A rule that accepted a patent on the agreement of several datasets would have accepted these with the highest confidence.

What is validated is therefore not the patent but the name: a statement of the form “this assignee name denotes this firm, or a company it controls, from this year to that year”. There are millions of patents and a few tens of thousands of names, a wrong attribution is almost always a wrong name, and once a name is validated and dated, every patent filed under it, at any office, follows mechanically. The names are read from Google’s table of patent publications of all offices (IFI CLAIMS Patent Services and Google 2026), from the printed assignee of a grant rather than from Google’s harmonised column, which for United States patents lists the inventors beside the company, and they are normalised with the same procedure as in Section 2, which removes the legal forms that patents write in full and reduces accented letters.

From a bridge’s patent to a validated, dated name

A bridge says that a patent belongs to a **gvkey**. We read the date of the link from the patent’s filing year, so that the datasets are put on one footing, and the crosswalk then says which FactSet entity holds that record in that year; a link is kept only if an entity does, which removes patents filed before the firms’ spans and patents filed in years in which the record belonged, for FactSet, to another entity. The printed assignees of the remaining patents, normalised, give the candidate pairs of a name and a firm. A pair is accepted by rule when the name matches, with the similarity and threshold of Section 2, a name that the record bore in the years in which the patents were filed, so that three conditions are met together: a bridge attributes the patents to the record, the firm holds the record in those years, and the company was called, in those years, what the patents call their owner. The Bosch patents cannot pass, since no name that Parker-Hannifin’s record ever bore resembles their assignee. The rule is weaker than that of the crosswalk in one respect, which we state: the bridges were themselves built by matching names, so that a bridge and a name match are in part the same evidence, and what this stage adds is the dates. As a guard against namesakes, a name that the rule would clear is sent to review when Google places the assignee mostly in another country than the firm’s.

The names that do not clear are subsidiaries, among them the companies that large firms create to hold their patents, companies acquired, joint ventures, co-assignees, former names, and the errors of the bridges. They were checked by the same combination of web searches and agentic reviews as the accounts (Section 2), each name beside the firm’s other names and records, to confirm or reject that the name denoted the firm, or a company it controlled, in the years its patents were filed. The question is: in the years these patents were filed, were patents filed under this assignee name the patents of this firm, of the company itself or of a company it controlled? The answer is a relation, the firm itself, a controlled subsidiary, a joint venture, a parent, a predecessor not owned, unrelated, or unsure, and the years in which it holds. Patents go up a group and never down: a subsidiary’s patents count for the firm that controls it from the year it controls it, and a parent’s patents never count for its subsidiary, so that a company acquired in 2009 brings the patents it files from 2009 and not those it filed before. The acceptance is decided in code from the relations, and an unsure is not accepted. What the review declined is the motivation of the stage: plain wrong attributions such as Bosch under Parker-Hannifin or Seagate under the former Symantec, the direction of ownership, LG Electronics’ patents under its subsidiary LG Display, and joint ventures that no partner controls, Fuji Xerox under Xerox. The years

matter as much as the relation: GlobalFoundries is a subsidiary of Advanced Micro Devices until its spin-off in 2009 and not after, and without an end year the patents of a divested business would stay with the seller, so that the outcome would be least responsive exactly where the paper looks for a response.

The product is a table with one row per name, firm and year, saying in which years a name denotes a firm. A cleared name is valid in the years in which the firm holds the record and the record bore the name, so that “Dow Chemical” is a name of Dow’s record until 2017 and not afterwards although the record continues. A reviewed name is valid in the years the review gave; a verdict without years holds from the first to the last filing year in view and no further, which is safe and can lose rightful years, so that the names for which patents were filed outside those years were dated by a further check of the years in which the name was the firm’s, or a company it controlled, with an explicit statement where the answer is “always”. Where a business passes from one owner to the next within a year, a year cannot settle whose the patents of that year are but a date can, and the names claimed by two firms in one year were dated to the day of the closing of the transaction; the panel carries those dates, and a patent is attached only if it was filed between them. A listed subsidiary beside its listed parent, VMware and EMC, Genentech and Roche, keeps its name in both, since each is a separate observation with its own consolidated perimeter, and the estimation clusters on the corporate group.

The names that no bridge shows

The bridges link United States patents to firms that Compustat and CRSP follow, and they leave gaps of three kinds: most firms outside the United States are not reached at all; a firm that is reached can have rightful names that no United States patent ever showed, a former name, a trade name, the name under which it files at home; and the bridges bring a firm’s subsidiaries only where a United States patent showed them. For all three the only route is the name, and it is a weak route, used narrowly. Every name we hold for a firm, FactSet’s names of every type, the Compustat names of its records and the Orbis names of the linked company, was looked up among Google’s assignee names of all offices, on a key that brings the abbreviations Google uses to one form, and a candidate required an equal key, the firm’s country where Google gives one, and filings in the years of the firm’s span. For the firms that report research and development and had no name, and later for every firm, the subsidiaries that FactSet’s ownership file (FactSet Research Systems 2021a) places under the firm were looked up the same way; that file is a picture of ownership today, without dates, and has errors, so that the review had to establish both that the assignee is the subsidiary and from when to when the firm controlled it. For the firms that report research and development and still had no name after these steps, a search of the web asked whether the firm patents at all and under which name, and the spellings it proposed were judged like the others. The unit judged is the spelling as Google harmonises it, with its country, because the normaliser makes distinct companies equal, and an equal name is where the search starts, not evidence: a spelling is accepted on a positive tie between the patents and the firm’s activity, size and research, and refused or left unsure otherwise. What is refused is what the name alone would have done: the patents of Huawei for a media company whose former name resembled it; owners, which the rule of direction excludes, Haier Smart Home for its listed subsidiary; ventures the firm does not control; and, the largest single group, the name of a later acquirer that Compustat wrote over the record of the company it bought, Canon over Axis, Renesas over Dialog. The finds are the operating companies of holdings, the transliterations under which a firm files at home, and former names.

Figure 21 shows the panel that results, by the way each name was validated. It holds 237,972 name-years for 27,514 names and 9,643 of the 18,216 focal firms (53%): 66,169 name-years from the names cleared by rule, 62,601 from the bridges’ names accepted on review, and 109,202 from the names found without a bridge. A firm has a median of 1 validated name and a mean of 3.0, the largest firm 238. A sample of the decisions of every kind was checked by hand, as for the crosswalk, and the checks found no wrong acceptance and no wrong refusal.

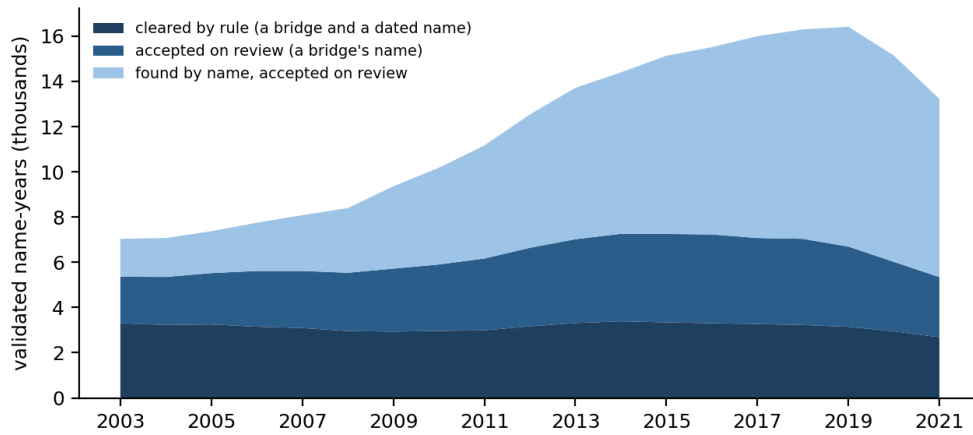


Figure 21: The dated name panel: validated name-years by year and by how the name was validated.

From names to patents

The patents are then taken from Google’s table of publications of all offices, in one pass, for every publication filed since 1999 whose assignee is a spelling of a list that holds every word-form of a validated name and every spelling accepted or still under review. Nothing is attributed by the pull. A publication is attached to a firm when one of its assignees meets a row of the name panel for the year in which it was filed, within the dates of control where the panel carries them, and never through a spelling that the review refused for that firm. A name validated on United States patents lets in spellings of every office, and the country polices them badly, since Google gives none for most filings at the Japanese and Chinese offices, so a guard uses families instead: a firm’s foreign filings share families with the patents it holds at home and a namesake’s share none, and a spelling of material volume whose families almost never contain a patent the firm holds through a home spelling was sent to review, which separated the namesakes, Systec K.K. of Japan under a firm whose Systec is American, from the foreign subsidiaries that file their own inventions locally. A namesake of the firm’s own country, filing at the same office, cannot be seen by this test or by any other we know of.

A firm that buys another buys its patents, and the count should measure what the firm invented. Two tests flag a family as acquired and both are recorded: the firm’s earliest filing in the family is more than twelve months after the family’s priority date, so that the invention existed before the firm held it; or the family’s earliest publication, at any office, names an organisation and none of the organisations it names is a name the firm holds. An empty assignee field and a person’s name are never evidence, the first being common on United States applications and the second an inventor. The outcomes then follow two units and two clocks. A granted patent is an application that received a grant, counted once in the year it was filed, at its office, the grant being read from the family members because at the Japanese, Korean and Chinese offices Google’s grant publications carry no assignee. A family is counted once, in its priority year, and marked originated or acquired. Applications never granted are kept, the kind of right (patent, utility model, design) is flagged and not dropped, and where two entities share a record in a year they are one reporting unit and the counts are united.

The panel

Table 6 gives the panel in numbers. Over 2003 to 2021 the firms filed 9,565,597 applications, of which 5,421,183 were granted, 2,082,744 of them at the United States office (38%); they hold 4,051,648 families with a priority date in the window, of which 3,791,533 originated with the firm and 260,115 (6.4%) are flagged acquired. 8,742 firms have at least one family in the window. Figure 22 follows the counts year by year: families originated rise from the middle of the period with the entry of the Asian firms into the

coverage, and the share of applications granted, steady at about 54% for filings up to 2015, falls for the last filing years to 59% in 2020 and 52% in 2021, because a grant lags a filing by two to four years and Google's table was loaded in 2026; the counts of the last years are lower bounds, and the estimation stops where the truncation begins. Figure 23 shows where the families are filed: 47% have a member at the United States office, 20% at the European office, and the share with a member at the Chinese office rises from 11% in 2003 to 41% in 2021, which is the composition of the firms rather than a change in their behaviour. Figure 24 gives the distributions: a firm-year with patents has a median of 5 families, a ninetieth percentile of 86 and a maximum of 13,610, and in 49% of the firm-years with patents no family is flagged acquired.

Table 6: The patent panel in numbers.

Validated assignee names	27,514
Name-years in the dated panel	237,972
Firms with at least one validated name	9,643
Patent applications filed 2003–2021	9,565,597
Of which granted	5,421,183
Of which granted at the United States office	2,082,744
Patent families with priority 2003–2021	4,051,648
Of which originated by the firm	3,791,533
Of which flagged acquired	260,115
Firms with at least one family, 2003–2021	8,742
Window firm-years with a validated name	70,724
Window firm-years with at least one family	40,536

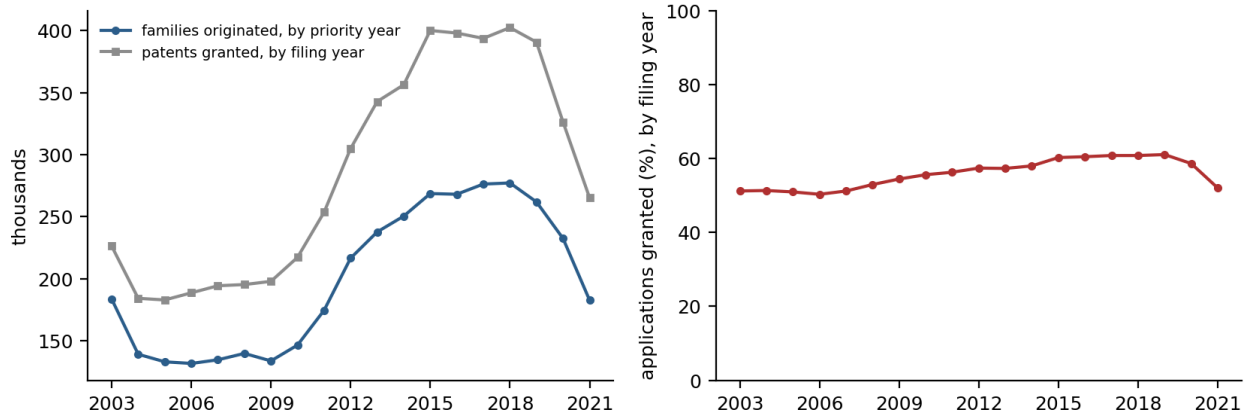


Figure 22: Left: families originated by priority year and patents granted by filing year, 2003–2021. Right: the share of applications granted, by filing year; the fall in the last years is truncation.

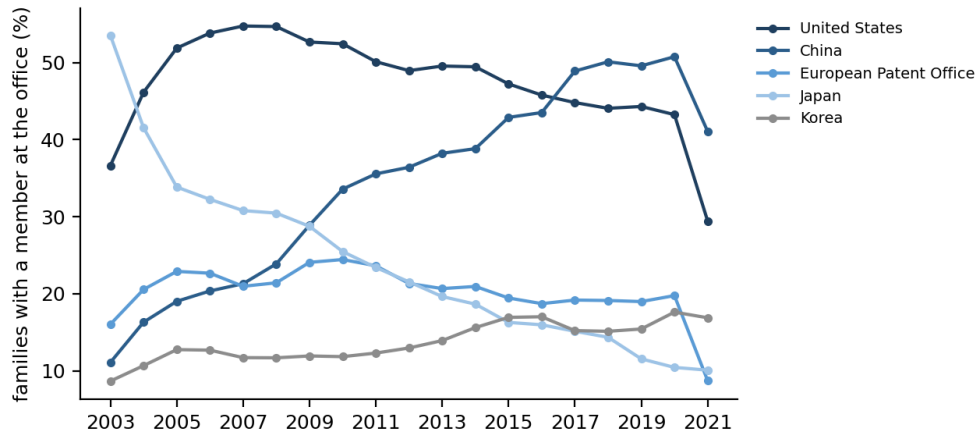


Figure 23: The share of originated families with a member at each of the five largest offices, by priority year.

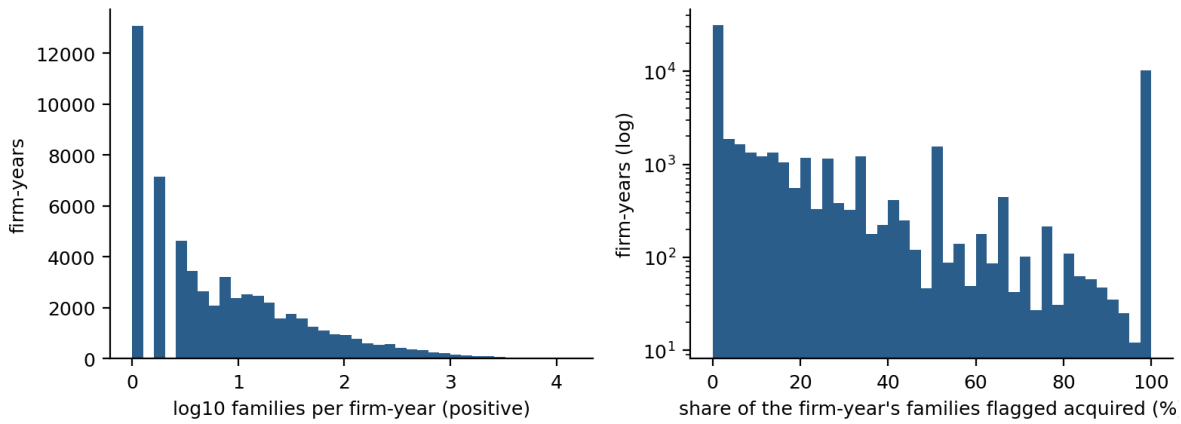


Figure 24: Left: families per firm-year over the firm-years with at least one, logarithmic scale. Right: the share of a firm-year's families flagged acquired.

What the panel covers, and what it does not, is stated in Figure 25. Of the 116,898 window firm-years, 61% belong to a firm with a validated name and 35% have at least one family in the year; the firm-years without a validated name, 8,610 firms, are recorded as missing and never as zero, since a firm without a name is a firm whose patents we cannot see, not a firm without patents. Weighted by the outcome the paper cares about the coverage is higher: 91% of the window firm-years with positive R&D belong to a firm with a validated name, and the 1,333 R&D-reporting firms without one are, on inspection of a random sample, mostly firms that file nothing under any recognisable name, software and services companies, shipping lines and biotechnology firms that license their compounds in, with a remainder of firms whose listed company is an offshore holding while an onshore operating company files, which no name we hold can reach. Two things bound the consequence. The estimates use firm fixed effects, so that a firm whose foreign subsidiaries are missed in every year is understated by a constant proportion that the fixed effect absorbs, and a firm without any patent contributes nothing; what would bias an estimate is an error that moves with mergers, a divested subsidiary kept or an acquirer's patents given to its target, and those are what the dating and the direction rule remove. One systematic cost of the direction rule should be stated: Chinese state groups file the patents of their listed operating companies in the name of the unlisted group, and those patents, being the parent's, are not the listed firm's.

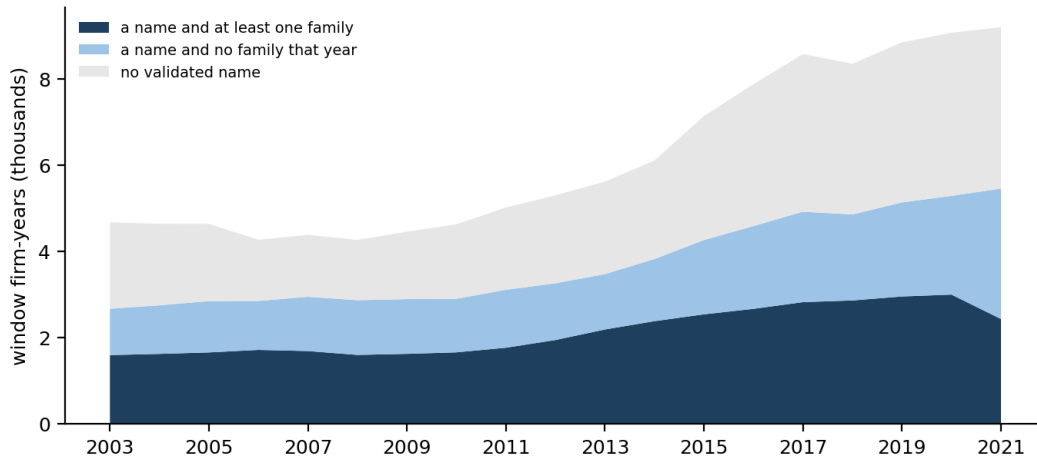


Figure 25: Window firm-years by fiscal year: with a validated assignee name and at least one family in the year, with a name and no family, and without a validated name (patents unknown).

For the families the firms originated, a last pull of Google’s table added the columns from which the quality and the direction of the inventions are read, and Table 7 describes them over the 59,796 firm-years with at least one originated family: the citations each family received within three and five years from families other than the firm’s own, the share of families cited within five years (68% of the families), the citations a family makes to earlier patents (7.6 per family) and to the scientific literature, the number of inventors (3.01 per family) and the share with an inventor abroad, the technology classes a family spans, and the families in a class the firm had not filed in during the previous five years (69,095 in all). The estimation takes the counts in logarithms or as inverse hyperbolic sines, the citation measures relative to the year and office of filing, and the class measures as the breadth and novelty of a firm’s research, each as an outcome placed in the fiscal period that contains the calendar year’s end.

Table 7: The patent outcome variables over the firm-years with at least one originated family, 2003–2021.

	firm-years	median	mean	p90
Originated families in the year	59,796	4.0	63.4	77.0
Forward citations within five years, per family	49,473	3.0	5.5	12.0
Share of families cited within five years, %	49,473	77	70	100
Backward citations per family	49,473	4.8	10.6	20.3
Non-patent literature citations per family	49,473	0.5	2.4	5.0
Inventors per family	49,473	2.7	3.0	5.0
CPC subclasses per family	49,473	1.5	1.6	3.0
Families in a subclass new to the firm	59,796	0.0	1.2	3.0
Share of families with a foreign inventor, %	49,473	0.0	22.7	87.5

4 The deals and the estimation panel

The treatment of the paper is a consolidation among a firm’s rivals: a merger or an acquisition in which a designated rival takes part, as the buyer, as the company bought, or, for the placebo and control classes, as the parent that sells a unit. The deals therefore have to be attributed not to the focal firms but to every company that is ever a rival of one of them, and, for the interference controls, to their suppliers, customers and partners as well: the whole universe of 212,674 entities of the rivalry network. This section explains where the deals come from, how each was attributed to the entity that took part in it, how a firm’s exposure in a year is then read from its rival set, and how the estimation panel is assembled and used.

The deals and their attribution

The deals are Refinitiv’s SDC Platinum mergers and acquisitions (LSEG Data & Analytics 2026*b*), read from WRDS: 1,348,776 deals announced from 1999 to 2026, 925,859 of them in 2003 to 2021, of which 80% were completed and 84% transfer control. Each deal names an acquiror, a target and their parents by name, nation and a six-digit issuer CUSIP, and the CUSIP is the only identifier. Two things make the attribution hard. The CUSIP is an identifier of a security, reassigned over time, and for companies that never had one the vendor states that it estimates one, so that a private company’s code meets nothing in our data and a match on it is never proof by itself. And the deal parties are not the same objects as FactSet’s entities: a deal is often recorded under a division, a subsidiary or a former name. The stage therefore does what the two previous sections did. It dates every identifier, building for each entity the intervals on which it held each CUSIP from four dated sources, the crosswalk of Section 2 with Compustat’s identifier history, FactSet’s dated chain from entity to security to CUSIP, CRSP’s dated CUSIPs, and Refinitiv’s own security master (LSEG Data & Analytics 2026*a*); it accepts by rule only when the identifier and the name agree on the day of the deal; and it sends the rest to a review that decides nothing by name alone.

One principle governs what is attributed: a deal is a firm’s when the firm itself is the party. Patents are an outcome and a group’s patents are consolidated up to the parent; a deal is the event that shocks the rivals of the firm that made it, and the paper asks whether the designated firm entered a consolidation, not whether some company in its group did. A subsidiary’s deal is therefore not its parent’s and a parent’s deal is not its subsidiary’s, for every kind of related firm alike, and a unit that SDC writes as “Owner-Unit”, a division or a plant sold out of a company, is refused as the owner and recorded as the owner’s divestiture instead. The one place the group is kept is free of judgement: when SDC records the acquiror as a vehicle it names the parent, and the parent counts as the actor in that case only.

Two routes propose candidates. On the identifier route the party’s CUSIP equals a spell of the entity that covers the announcement date, and the link is cleared by rule when the party’s name also matches a name of the entity and the party’s nation is the entity’s country; a CUSIP match whose name or nation disagrees, which is mostly a unit recorded under the firm’s code, “General Electric Capital Corp” under General Electric’s, goes to review. On the name route, for the entities without a CUSIP and the dates no spell covers, the party’s name equals a name of the entity and the deal is announced inside the calendar years the entity’s relationships touch, since a rival’s deal outside those years is never an exposure; one stratum is accepted by rule, the distinctive current name, a name of two words or more borne by exactly one entity of the whole universe, equal in a strict form that keeps word order, and every other candidate is checked by the same kind of review as the accounts (Section 2), because an equal name proves little for the small and private companies this route reaches. The review sets out to confirm or reject that the SDC party is the entity itself at the date of the deal, comparing what the entity does, its size and its names with SDC’s description of the party and its deals; a party confirmed for two entities was given to one claimant per range of years.

Table 8 and Figure 26 give the result: 484,628 links between an entity, a deal and a role, for 59,079 entities and 377,494 deals, 65% of them by the identifier rule, 18% by the distinctive-name rule and 18% by review. 15,223 focal firms and 22,467 named rivals (30% of them) have at least one link. Every link carries the deal’s flags as facts of the record, its outcome, whether it transfers control, whether it is a buyback or a recapitalisation, whether both sides belong to one group by SDC’s parents, by one entity, by one reporting unit or by the acquiror’s own list of subsidiaries in its annual report (Wharton Research Data Services 2026*b*), and what counts as a consolidation is decided from these flags at estimation, in the open. A hand check of every kind of link, drawn with a fixed seed, agreed with the identifier rule and the distinctive-name rule in all but one row each, and with the review in all but a few rows where the review had accepted a holding for its operating layer.

Table 8: The (entity, deal, role) links by the way they were attributed and by role.

	identifier rule	distinctive-name rule	review	all
as the buyer	124,778	35,269	56,678	216,725
as the company bought	38,534	20,447	26,289	85,270
as the seller of a unit	86,146	30,250	2,983	119,379
as the buyer's parent (vehicle deals only)	63,254	0	0	63,254
all roles	312,712	85,966	85,950	484,628

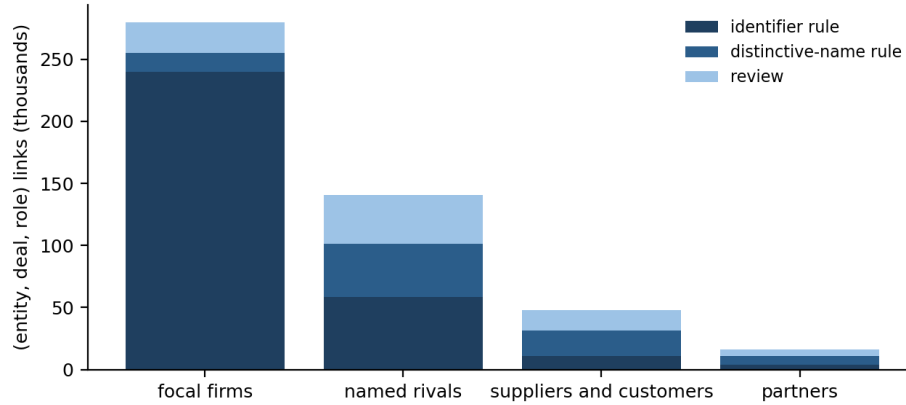


Figure 26: Links by the way they were attributed, for the focal firms, their named rivals, their suppliers and customers, and their partners.

The events of the rivals

Every link receives a class by rule from the deal's flags. A consolidation is a completed control-changing deal, dated at its announcement, that is neither a buyback nor within one group by any of the four tests: the entity is bought, or the entity buys control. A divestiture is a unit sold; a partial deal is a minority stake or an asset purchase; a withdrawn deal is a control deal announced and withdrawn; rumours and pending deals are kept and never a treatment, a rumour being read from the deal's outcome and not from SDC's flag, since many completed deals began as rumours. Over 2003 to 2021 the named rivals of the focal firms took part in 101,308 consolidations, 87,847 as the buyer and 13,461 as the company bought, and 25,534 rivals (34% of all named rivals) consolidated at least once; they sold a unit 51,176 times and saw 4,144 control deals withdrawn (Figure 27). A value is disclosed for 50% of the consolidations, with a median of 78 million dollars and a ninetieth percentile of 1,298 million; 35% cross a border and in 13% the company bought is listed.

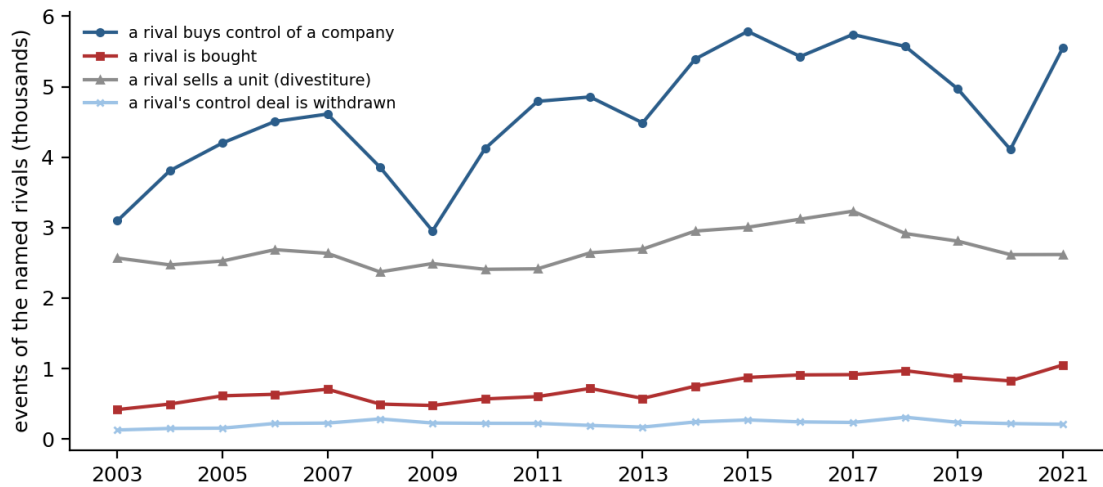


Figure 27: Events of the named rivals of the focal firms by announcement year, 2003–2021.

A check of the other direction tests the target side, which the paper relies on most. Compustat records the date on which it stopped following a company and the reason. Of the 3,421 focal firms that Compustat deleted inside their window for an acquisition or a merger, our own target-side consolidation event lies within two years of the deletion for 2,915 (85%); the disagreements, read one by one, are a parent buying out its listed subsidiary, which is within one group and not a consolidation of rivals, a merger in which Compustat kept the other company’s record, a stake below full ownership with the firm still listed, and acquisitions that SDC filed under a variant of the issuer’s code that no source gives us, which the name route proposes and the review judges. In the other direction, 1,107 firms with our event stay in Compustat two years on, a majority acquisition of a company that remains listed.

From the rivals’ deals to a firm’s exposure

For each focal firm and fiscal year the rival pair-years of Section 1 are joined to the rivals’ events, and the year’s exposure is counted by class: the distinct events, the distinct rivals with an event, their share of the year’s live rival set, and the deal value. A deal counts in the fiscal period of the firm that contains its announcement date, and a firm-year without accounts takes the calendar year. The same counts are made with the event required to fall inside the record’s own dates, the exact reading of Section 1, over the undirected and the mutual sets, over the rivals of rivals, over the suppliers and customers and over the partners, for the firm’s own deals, and for the placebo classes. The distance between the two headquarters travels with every rival event, so that the exposure can be weighted or split by distance. And the case in which a rival buys a company that is also a rival of the same firm that year, the horizontal consolidation, is counted separately.

From these counts the estimation reads its shocks, and Table 9 and Figure 28 give them on the estimation sample under the calendar reading. In 74% of the estimation firm-years at least one designated rival takes part in a consolidation, which is an intensity rather than an event: an exposed firm sees a median of 6 deals by 3 rivals in the year, and 33% of its rival set consolidating (Figure 29). The paper’s shocks are therefore the events inside this intensity. A rival is bought in 21% of the firm-years, for 8,925 firms. At least half of the live rival set consolidates, the threshold shock, in 21%. A rival buys another rival of the same firm, the horizontal shock, in 6%, for 4,040 firms. A rival’s control deal is withdrawn or remains a rumour, the placebo, in 38%. The share of exposed firm-years is highest in the first years, which are the years in which the coverage was North American and the rival sets those of large listed companies, and it falls as the smaller firms of the other regions enter.

Table 9: The exposure classes over the estimation firm-years, fiscal 2003–2022, calendar reading of the relationships.

shock, calendar reading, estimation firm-years 2002–2022	firm-years	share	firms
any designated rival in a completed control deal, either role	78,223	74%	15,305
a designated rival buys control of a company	76,674	73%	15,103
a designated rival is bought	22,194	21%	8,925
at least half of the live rival set in such deals (the threshold shock)	22,321	21%	7,943
a rival buys another rival of the same firm (the horizontal shock)	6,664	6%	4,040
the same over the undirected set (rivals and namers)	84,503	80%	15,624
the same over the mutual rivals	24,474	23%	6,210
a rival of a rival in such a deal	80,770	76%	15,112
a supplier or customer in such a deal	73,106	69%	13,739
a partner in such a deal	55,021	52%	11,673
a rival's control deal withdrawn or a rumour (placebo)	40,166	38%	11,058
the firm itself in such a deal	24,672	23%	8,350

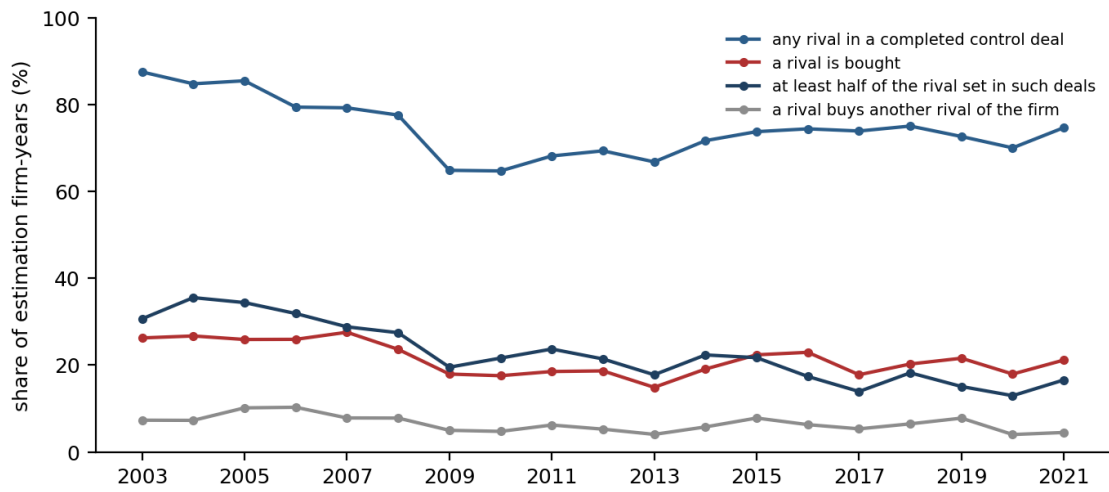


Figure 28: The share of estimation firm-years exposed, by fiscal year and by shock.

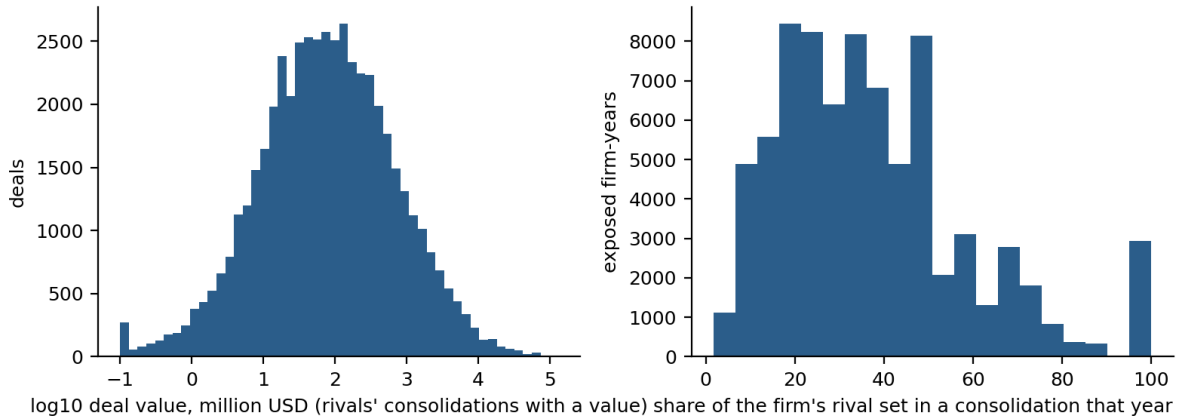


Figure 29: Left: the value of the rivals' consolidations, where disclosed, in millions of dollars, logarithmic scale. Right: the share of a firm's live rival set taking part in a consolidation in the year, over the exposed firm-years.

The estimation panel

The panel has one row per focal firm and fiscal year, the fiscal year following Compustat's convention, over the fiscal years the firm's window covers and the two periods before it, which supply lags only. Each row carries the accounts and the R&D of Section 2 with their status and level, the patents of Section 3

with the missing flag and the distance to the data’s last year, the exposures above under each reading of the relationships, the firm’s own deals, the year founded and the industry, and the reporting unit, so that two focal entities that share one record in a year are one cluster and the second is flagged as a duplicate. Two rules of Section 1 carry into it. Before the firm’s first record as a source firm every exposure is missing, since its rival set is unknown. Inside its coverage, a year in which the firm has no live rival record is a year in which its documents named no competitor, and the analysis treats it in two ways and reports both: as zero exposure with an indicator that enters at that year, or as a missing year. The two readings of the relationship dates, calendar and gap-filled, and the two treatments of a silent year give four versions of every exposure, and every result of the paper is run on each.

Table 10 counts the panel down to the estimation sample. Of the 117,613 window firm-years, 105,895 carry the firm’s own accounts and 105,678 of those fall inside the firm’s coverage; this is the estimation sample, for 17,361 firms, and R&D is observed in 58,548 of its firm-years, positive in 51,726, for 9,902 firms. A covered year without a live rival occurs in 7,697 firm-years of 1,876 firms. The main analysis uses the firm’s own accounts and drops the duplicate entity of a reporting unit; the sample that adds the buyer’s consolidated accounts to a bought firm is a variant.

Table 10: From the window firm-years to the estimation sample.

	firm-years	firms
Window firm-years, fiscal 2002–2022 (the firm names a competitor)	117,613	18,205
With accounts of any kind	107,598	17,450
With the firm’s own accounts	105,895	17,377
With own accounts, inside the firm’s coverage (the estimation sample)	105,678	17,361
Of which with R&D observed (positive or a zero)	58,548	9,902
Of which with positive R&D	51,726	8,884
Of which a covered year without a live rival (the reporting dummy)	7,697	1,876
Of which the second entity of a reporting unit (dropped)	599	283

How the panel is used is set out in the main text, and only the outline belongs here. The binary designs read the yearly shock indicator with a local-projections difference-in-differences estimator, regressing the change of the outcome from the year before the shock to each horizon on the indicator among clean observations, with year effects within industry, in three readings of the indicator: a firm’s first exposure ever, with the not-yet-exposed firms as controls; a shock after a quiet spell of three or five years, kept at a horizon only if no further shock arrives, with the firms whose status does not change over the window as controls; and the same shock read as the start of a run of exposure. Each reading is estimated for the designated, the undirected and the mutual rival sets, for the any-deal, the threshold and the horizontal shock, on a ladder of specifications that adds the controls at the year before the shock, excludes the years around the firm’s own deals, conditions on the firm’s growth, and adds the deals of the namers, of the rivals of rivals, of the suppliers and customers and of the partners as interfering shocks, together with a dummy for a wave of deals in the firm’s industry. The placebo classes, the withdrawn deals and rumours, the industry wave, and three random rival sets drawn with replacement from the entities that are someone’s rival in the year, are run through the same designs. The continuous design replaces the indicator with the share of the rival set consolidating in the year and uses the intertemporal estimator of de Chaisemartin and D’Haultfœuille, and the patent outcomes enter the same designs with the truncated years dropped.

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